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Main group chemistry: from molecules to materials

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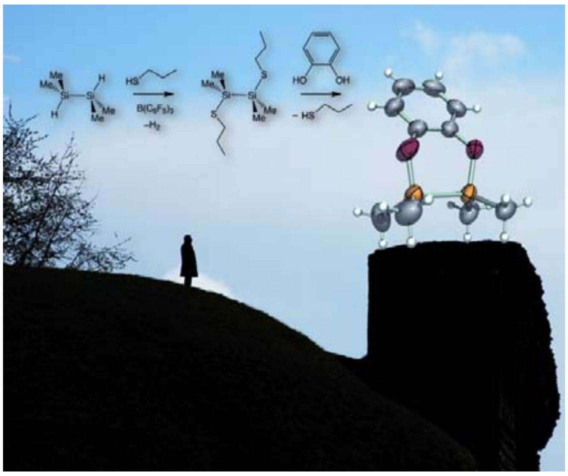

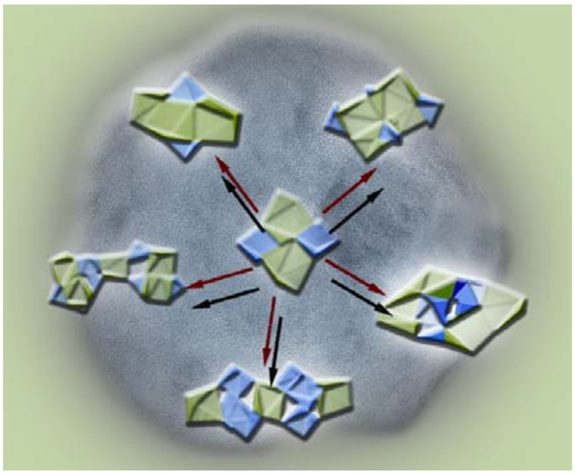

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Hydride encapsulation by molecular alkali-metal clusters

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The sequential treatment of group 12 and 13 Lewis acids with alkali-metal organometallics is well established to yield so-called ''ate' complexes, whereby the Lewis-acid metal undergoes nucleophilic attack to give an anion, at least one group 1 metal acting to counter this charge. However, an alternative, less well recognised, reaction pathway involves the Lewis acid abstracting hydride from the organolithium reagent *via* a β-elimination mechanism. It has recently been shown that in the presence of N,N'-bidentate ligands this chemistry can be harnessed to yield a new type of molecular main-group metal cluster in which the abstracted LiH is effectively trapped, with the hydride ion occupying an interstitial site in the cluster core. Discussion focuses on the development of this field, detailing advances in our understanding of the roles of Lewis acid, organolithium, and amine substrates in the syntheses of these compounds. Structure-types are discussed, as are efforts to manipulate cluster geometry and composition as well as hydride-coordination. Embryonic mechanistic studies are reported, as well as attempts to generate hydride-encapsulation clusters under catalytic control.

General introduction

The last two decades of the 20th century saw massive advances in the development of generic structural principles governing organolithium compounds and their relatives, lithium amides, imides and the like.[1] These principles, which led to lithiated organics achieving unparalleled status in terms of our structural understanding of a family of metalloorganic species,[2] were seminally expressed by, amongst others, the late Ron Snaith in terms of (LiX) ring formation through the structure-defining effects of electrostatic interactions, and the subsequent association of these metallocycles through (vertical) stacking[3] or (lateral) laddering events.[4] The comprehensive success of ring stacking and laddering theories in rationalizing the structural motifs observed for organolithium compounds also suggests an explanation for the lack of a polyhedral structural chemistry for these species. While these cluster architectures, frequently noted for transition-metal clusters and capable of exhibiting anion recognition and encapsulation properties, have been reported on occasions, they are far from commonplace.

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Embedded ions: redefining structural principles for the group 1 metals

Anion entrapment by two-dimensional alkali-metal arrays has focussed on the development of mixed group 1–group 2 metallocycles. Noted originally in 1998,[5] eight-atom heterometal architectures of the type $(MNM'N)_2$ (M = alkali metal, M' = alkaline-earth metal) have revealed the flexibility to coordinate both oxide and peroxide within the same lattice[6] and also to trap two equivalents of the alcoholate $C_8H_{17}O^-$.[7] Related chemistry has yielded the larger macrocyclic system $[(tmp)_6Mg_2Na_4]^{2-}$ **1**$^{2-}$ (tmp = 2,2,6,6-tetramethylpiperidide) whose cavity has proved

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capable of incorporating a 1,4-deprotonated benzene dianion to give $(tmp)_6(C_6H_4)Mg_2Na_4$ **1**.[8] More recently this phenomenon has been repeated using mixed s-/d-block chemistry.[9] Nonetheless, the vast majority of documented three-dimensional alkali-metal architectures (that can be more meaningfully regarded as encapsulation polyhedra) are oxide-wrapping species of the type $M_{n+2}O(R)_n$.[10] These most commonly reveal μ_6-coordination of an oxide and are based on an octahedral $[Li_6O]^{4+}$ motif, as in $\{(c\text{-}C_5H_9)N(H)\}_{12}OLi_{14}$ **2** or the fused bis(pseudo-cubane) architecture of $(salen)_3OLi_8{\cdot}2tmeda$ **3** [tmeda = *N*,*N*,*N*′,*N*′-tetramethylethylenediamine; salen = *N*,*N*-ethylenebis(salicylideneimide) dianion] (Fig. 1).[11,12] Indeed, the role of ligand structure in defining polyhedral motifs has been highlighted, with *N*,*N*′-bidentate ligands enabling the formation of a cage ideal for oxide encapsulation with μ_8-coordination of the interstitial ion noted. Several isostructural examples are known, incorporating near cubic metal arrays with face-capping anions based on 4-azabenzimidazole,[13] *N*,*N*′-bis(cyclohexyl)formamidine,[14] or 1,3,4,6,7,8-hexahydro-2*H*-pyrimido[1,2-*a*]pyrimidine (= hppH).[15] A recent and intriguing extension of this area of anion encapsulation utilising the last of these highly symmetrical *N*,*N*′-bidentate ligands involves the isolation and characterisation of μ_9-peroxide encapsulation cluster $(hpp)_7(O_2)Li_9{\cdot}n$thf (n = 0, 1) **4** (Fig. 2).[16] A recent DFT geometry optimisation (B3LYP method) of this structure (with n = 0) reinforces the view of Coles and Hitchcock that the complex incorporates an embedded peroxide dianion.[17] Other homometallic peroxide capture species have been reported, including the products achieved if dilithium *N*,*N*′-bis(trimethylsilyl)hydrazide is hydrolysed[18] and if MeLi is reacted with 1,3-dimethyl-2-iminoimidazoline,[19] whilst intermetallic Li–Rb clusters have also trapped peroxide.[20] In essence, all these structures incorporate M_2O or M_2O_2 fragments, supported by a metal amide superstructure. Whilst fundamentally altering the nature of this encapsulating framework has been demonstrated

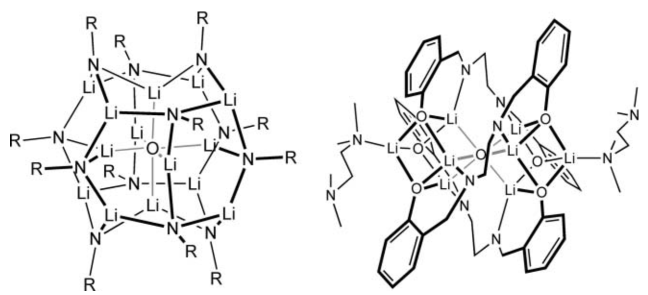


Fig. 1 Octahedrally disposed oxides of lithium: $(RNH)_{12}OLi_{14}$ **2** (R = c-C_5H_9) and $(salen)_3OLi_8{\cdot}2tmeda$ **3**.[11,12]

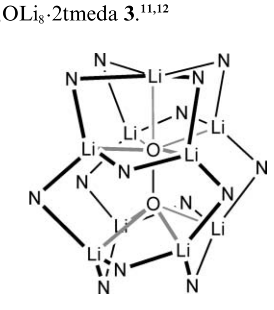


Fig. 2 Core of the thf-free version of peroxide capture cluster $(hpp)_7(O_2)Li_9$ **4**.[16]

to significantly influence interstitial anion coordination, data on the manipulation of anion bonding in chemically and structurally analogous frameworks and, indeed, on the formation of anion-free ‘parent’ clusters is much harder to come by. That said, investigations into the structures and reactivities exhibited by dilithium (silyl)phosphanediides and arsanediides have shown that, empirically, H_2ER_3 [E = P, R = $Si(CMe_2{}^iPr)Me_2$; E = As, R = $Si(Mes)^iPr_2$ (Mes = mesityl)] reacts with $^nBuLi/Li_2O$ to give octameric and dodecameric aggregates, respectively.[21] The first of these products, $(PR_3)_8OLi_{18}$ **5**, incorporates an octahedral $[Li_6O]^{4+}$ motif, and efforts to establish whether this core acts as a nucleating agent about which a cluster shell can aggregate have led to the synthesis of $(PSi^iPr_3)_{10}Li_6$ **6** in the absence of Li_2O.[22] It has also been noted that the reaction of $H_2PSi(CMe_2{}^iPr)Me_2$ (= H_2PR) with ^nBuLi in the presence of trace LiOH in toluene affords a partially lithiated dodecameric species $(PR)_6(HPR)_6OLi_{20}$ **7** (Fig. 3(a)) which reveals an “open” three-dimensional, cyclized $P_{12}Li_{18}$ ladder encapsulating a structure-determining, linear Li_2O unit. The same study has established that this represents a precursor to formation of the fully-metallated μ_6-oxide complex $(PR)_{12}OLi_{26}$ **8** (Fig. 3(b)).[22]

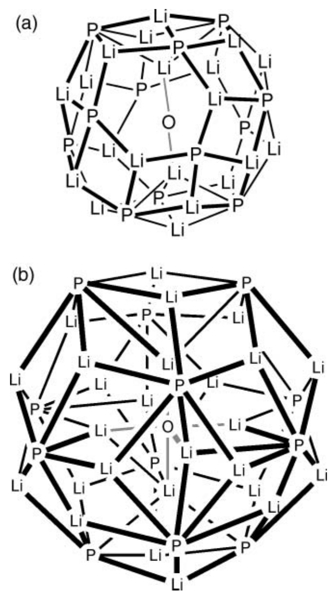


Fig. 3 (a) The “open” core of partially lithiated μ_2-oxide $(PR)_6(HPR)_6OLi_{20}$ **7**. (b) Complete metallation yields the μ_6-oxide $(PR)_{12}OLi_{26}$ **8**.[22]

Whereas the vast majority of species incorporating embedded anions (hydrides notwithstanding, *vide infra*) are oxides, halide entrapment has also been recorded. The condensation reaction of $\{H_2NP(\mu\text{-}N^tBu)\}_2$ with $\{ClP(\mu\text{-}N^tBu)\}_2$ in the presence of Et_3N is known to yield the cyclic tetramer $[\{P(\mu\text{-}N^tBu)\}_2(\mu\text{-}NH)]_4$ **9**.[23] However, it has more recently been shown that the presence of lithium halides templates formation of the pentamer $[\{P(\mu\text{-}N^tBu)\}_2(\mu\text{-}NH)]_5X^-$ (X = Cl, Br, I) **10** with distortion of the pentameric framework and thermodynamic stability of the system increasing with guest size and the halides acting as kinetic templates for formation of the pentamer over that of **9**.[24] An extension of this work has recently seen the coordination of LiCl by a group 13 phosphide host, the reaction of $MeECl_2$ with ‘$PhPLi_2$’ in thf giving $[\{MeE(PPh_3)Li_4{\cdot}3thf\}_4(\mu_4\text{-}Cl)]Li{\cdot}n$thf (E = Al, n = 0 **11a**; E = Ga, In, n = 3 **11b**).[25] Lastly, a single example of trigonal bipyramidally encapsulated fluoride in $\{(Me_3SiN)_2CC_6F_5\}_4(F)Li_5$ **12** has also been reported (Fig. 4).[26]

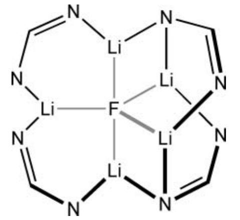


Fig. 4 Core of trigonal bipyramidal $\{(Me_3SiN)_2CC_6F_5\}_4(F)Li_5$ **12**.[26]

Hydride capture and encapsulation by transition-metal systems

While oxide and peroxide inclusion, sometimes targeted and sometimes serendipitous, has come to dominate examples of alkali-metal polyhedral clusters, there is more general interest in cages capable of acting as hosts to other small ions. One category of embedded ion of particular novelty is the interstitial hydride. Previous examples in this field are few and far between and tend to involve transition metals.

Trigonal μ_3-hydride coordination has been reported for mixed Li–Sm[27] and Li–Zr[28] systems while a tetrahedral environment has been suggested in bridged zirconium hydrides.[29] Tetrahedral coordination has also been proposed for the hydride ion embedded in a Nb_6 cluster.[30] More recently the existence of μ_4-hydride has been unambiguously established in $Y_4H_8(Cp')_4$ **13** $\{Cp' = C_5Me_4(SiMe_3)\}$ (Fig. 5).[31] The ruthenium cluster $[Rh_{13}H_2(CO)_{24}]^{3-}$ **14**$^{3-}$ was first identified in 1975[32] but it is only recently that neutron analysis has demonstrated that both hydrides are μ_5-coordinated with the anions displaced slightly towards the core of the cluster (H–Rh(core) 1.84(2) Å, H–Rh(surface) 1.97(2) Å) (Fig. 6).[33] Octahedral encapsulation has proved to be a dominant motif in related systems. The first single-crystal neutron diffraction data collection to indicate this revealed regular octahedral hydride coordination in both $[Ru_6H(CO)_{18}]^-$ **15**$^-$ (mean H–Ru 2.038 Å, range 2.028(11)–2.055(12) Å)[34] and $[Co_6H(CO)_{15}]^-$ **16**$^-$ (mean H–Co 1.823 Å, range 1.799(17)–1.872(16) Å)[35] (Fig. 7) and, subsequent to this, methods for hydride location were summarized.[36]

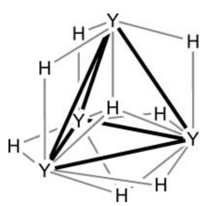


Fig. 5 The core of $Y_4H_8(Cp')_4$ **13** $\{Cp' = C_5Me_4(SiMe_3)\}$ as revealed by recent neutron diffraction analysis.[31]

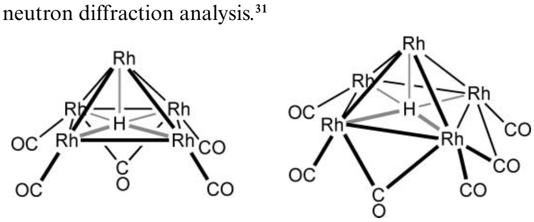


Fig. 6 Structures of the environments around the five-coordinate hydride ions noted in the neutron-diffraction structure of $[Rh_{13}H_2(CO)_{24}]^{3-}$ **14**$^{3-}$.[33]

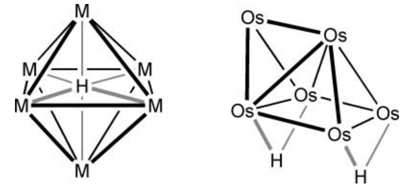


Fig. 7 The neutron structures of $[M_6H(CO)_n]^-$ (M = Ru, n = 18, **15**$^-$; M = Co, n = 15, **16**$^-$)[34,35] reveal classic octahedrally bound interstitial hydrides. However, in spite of the existence of an interstitial vacancy, the hydride ions in $Os_6(H)_2(CO)_{18}$ **18** were proposed to exist at the cluster surface on the basis of an X-ray diffraction structure,[37] and this has been borne out by neutron crystallography.[38]

The analysis of Os clusters has proved more protracted. Early work led to isolation of the anion $[Os_6H(CO)_{18}]^-$ **17**$^-$ and of neutral $Os_6(H)_2(CO)_{18}$ **18**. The former adopted a classic octahedral structure, with the expansion of one Os_3 face being attributed to the presence of a face-capping hydride.[37] The same work used a combination of X-ray diffraction analysis and NMR spectroscopy to infer the presence of μ_2-hydride ions bridging Os–Os edges in the unconventional capped square pyramidal structure of **18**.[37] This view, expounded by McPartlin *et al.*, was subsequently corroborated by single-crystal neutron diffraction analysis (Fig. 7) and led to the conclusion that "Just because a cavity is empty does not necessarily mean that a H atom will automatically fill it".[38] The extensive cluster $[Os_{10}H_4(CO)_{24}]^{2-}$ **19**$^{2-}$ has been characterized by neutron diffraction, and the four hydrides found to reside at the surface of the metal framework; two μ_2 (edge-bridging) and two μ_3 (face-capping).[39] These motifs have also been noted in conjunction with the encapsulation of interstitial hydride in **13**.[31] Lastly, spectroscopic evidence has been provided for edge–terminal–edge hydride scrambling in the cluster $Ru_4H_4(CO)_{10}(dppe)$ **20**.[40]

Main-group hydrides

Turning to main-group systems, chemical and physical aspects of main-group hydrides have been reviewed.[41] However, only a handful of molecular s-block hydrides have been fully structurally authenticated. Prior to our work the full structural parameters of only a single lithium hydride cluster, $(^tBuO)_{16}H_{17}Li_{33}$ **21**, were known (Fig. 8).[42] This species formed reproducibly if $^tBuOLi/^tBuLi$ mixtures were photolysed, and Fourier difference synthesis has allowed both the location and isotropic refinement of octahedrally bonded hydride ions within the cluster core and pentagonally coordinated hydride at the cluster surface (H–Li range 1.81(4)–2.24(4) Å). Efforts directed towards the storage of hydride have led to the development of small interstitial intermetallic hydrides.[43,44] Moreover, extensive advances have been made in the development of complex hydrides, for example alanates (*e.g.*, $NaAlH_4$ **22**), as hydrogen carriers and the field has been recently reviewed.[45] These have been found to release hydrogen through decomposition/recombination processes (*e.g.*, **22** $\Leftrightarrow \frac{1}{3}Na_3AlH_6 + \frac{2}{3}Al + H_2 \Leftrightarrow NaH + Al + \frac{3}{2}H_2$) that, in the presence of metal catalysts (*e.g.*, Ti, Zr), point to new prospects for gas storage media.[46] There are many examples known of lithium aluminates that incorporate Al(μ-H)Li motifs,[47] but just two in which embedded hydride has been noted. Recently, a highly distorted $AlLi_3$ environment has been noted for hydride

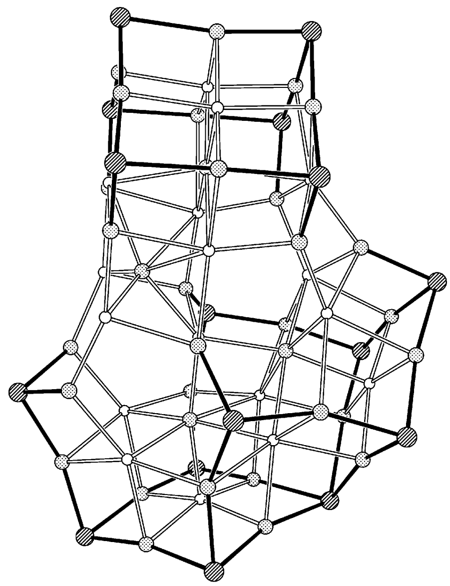


Fig. 8 Core of the non-trivial polyhydride $({}^tBuO)_{16}H_{17}Li_{33}$ **21**.[42]

in the pseudocubane $(hmdsAlH_2)_3HLi_4$ **23** (hmds = hexamethyldisilamide, Fig. 9). Distortion of the cluster, resulting from the insertion of H–Al units into three edges of the core Li_4H_4 cube, results in bonding H–Li distances lying in the range 1.82(1)–2.26(2) Å, with the final Li centre in the immediate vicinity being extruded and the embedded hydride being rendered pseudo-trigonal.[48] A similar pattern is noted if this species is part hydrolysed, with only the exposed μ_3-hydride undergoing reaction during the formation of $(hmdsAlH_2)_3(OH)Li_4$ **24** (Fig. 9).[48] Free refinement of the interstitial anion in $\{({}^tBuCH_2O)_5Al_3H_5\}Li{\cdot}OEt_2$ **25** has revealed a tetrahedrally coordinated hydride within the distorted Al_3Li cluster core (H–Li 2.00(3) Å, Fig. 9).[49] For higher group 1 metals in conjunction with Mg, the inverse-crown analogue $[({}^iPr_2NMgM)_2]^{2-}$ **26**$^{2-}$ has been found to encapsulate not oxide or peroxide dianions but two μ_2-hydride ions (H–M 2.68(3) Å for M = Na·PhMe **26a**, 2.96(3) Å for M = K·PhMe **26b**; Fig. 10).[50–52]

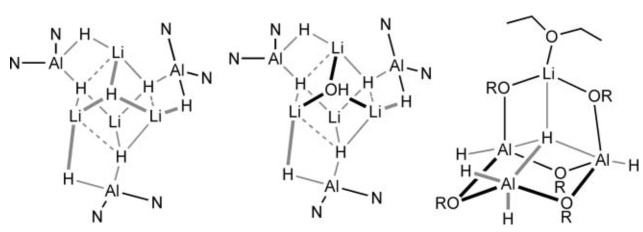


Fig. 9 Molecular structures of $(hmdsAlH_2)_3HLi_4$ **23**, $(hmdsAlH_2)_3(OH)Li_4$ **24** (left and middle, N = hmds) and $[(RO)_5Al_3H_5]Li{\cdot}OEt_2$ **25** (R = ${}^tBuCH_2$) (right).[48,49]

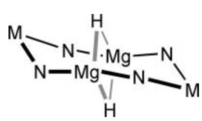


Fig. 10 Core of $({}^iPr_2NMgMH)_2$ (M = Na·PhMe **26a**, K·PhMe **26b**).[50,51]

An 'anion-free' cage: a polyhedral chemistry for lithium?

As the opening to this article sets out, advances in our understanding of lithium amide structural chemistry have led to the view that electrostatically-driven metallocycle formation gives the most basic building blocks and that these may or may not then undergo aggregation. Where they do, lateral association prevails and cyclized[53] or sinusoidal[54] ladders result. Importantly, any deviations from this principle appear to require the templating effect of some ulterior charge carrier. This may manifest itself in a variety of forms (*e.g.* co-crystallization of a second metalloorganic species).[7–9] However, the achievement of 3D lithium amide cages has hitherto apparently had two prerequisites. The first is that the ligand structure of the lipophilically wrapping amide anions must allow them to support the integrity of the otherwise non-bonded metal array. The second being that anion encapsulation must provide an attractive domain of negative charge density at the cluster core, overcoming inter-metal repulsion.[10,14–17,26] It appeared that a 3D polyhedral architecture for the alkali metals would not be achievable unless both criteria were met.[10] The requirement that, in contrast to transition-metal cluster chemistry, the exterior ligands are wholly responsible for defining the positions of the alkali-metal ions has meant that bidentate ligands with a bite angle large enough to disfavour chelation have usually,[13–15] though not invariably,[11] been required. The tendency for alkali metals to undergo chelation in the presence of flexible bidentate ligands (tmeda, dme *etc.*) has introduced the further need for ligand rigidity and, in the context of *N*,*N*′-bidentate ligands, this has been done by linking the donor centres with an sp^2 geometry carbon atom. Nevertheless, the requirement for interstitial electron density has meant that, although oxide and peroxide encompassing structures have been recorded, unoccupied analogues—effectively the parent clusters—have hitherto gone unreported; either isolable material proved unobtainable or else it revealed very different structure types. This changed in 2003 when we reported the synthesis and structure of the $[(hpp)_6Li_8]^{2+}$ **27**$^{2+}$ dication (Fig. 11, the core of which is given in Chart 1).[55]

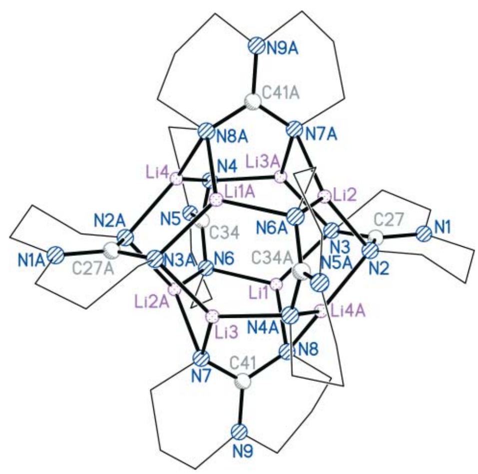


Fig. 11 X-Ray crystal structure of the cubic dication component of $\{[({}^tBu_2AlMe_2)_2Li]^-\}_2[(hpp)_6Li_8]^{2+}$ **27**.[55] As for all diffraction structures, minor disorder and H-atoms (interstitial hydride notwithstanding, where appropriate) are omitted for clarity.

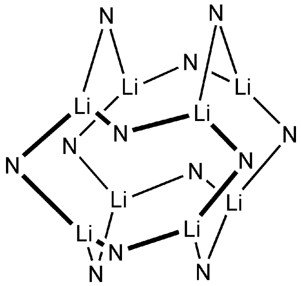
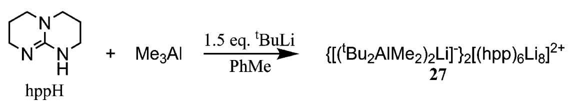


Chart 1

The route by which this ion is obtained is, in essence, the same as that outlined below for the encapsulation of interstitial hydride, and it is unclear whether the formation of **27**$^{2+}$ occurs in place of hydride encapsulation or whether this species represents one component of a more complex mixture. To date, however, no evidence has been found for co-formation of the corresponding hydride-containing monocation. The combination of hppH with equimolar $AlMe_3$ (known to afford the dimethylaluminium(guanidinate) (hpp)$AlMe_2$ **28**)[56] was followed by treatment with 1.5 eq. ^tBuLi to yield a single isolable crystalline product (Scheme 1). The use of excess organolithium is discussed in detail in the next section. Indeed, the original premise of our work on mixed Li–Al systems was the study of simple alkali-metal aluminates, and for such a species ^1H NMR spectroscopy should reveal a straightforward 1 : 1 : 1 ratio for hpp, ^tBu and AlMe fragments here. However, the observation of a 3 : 2 : 2 ratio pointed to more complex reaction and X-ray crystallography reveals ion-separated $\{[(^tBu_2AlMe_2)_2Li]^-\}_2[(hpp)_6Li_8]^{2+}$ **27** (where **27** = $(\mathbf{27}^-)_2\mathbf{27}^{2+}$, Fig. 11 and 12) in a yield of 57% (computed on the basis that ^tBuLi is the limiting reagent according to the crystallographically determined identity of the product).[55] The $[(^tBu_2AlMe_2)_2Li]^-$ **27**$^-$ ions contain two tetraalkylaluminate centres arranged with two methyl substituents on each, tetrahedrally wrapping the central Li^+ ion. This motif, unusual in itself, proves to be a recurrent feature in this area of study. Although the action of the bridging alkyl groups is replicated in the polymer of $LiAlEt_4$ **29**,[57] molecular congeners are hard to come by. Few examples of alkali metal-containing anions are known,[58] the closest analogue



Scheme 1

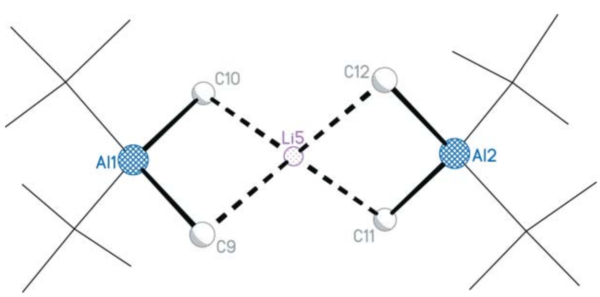


Fig. 12 The anionic lithium bis(aluminate) component of ion-separated $\{[(^tBu_2AlMe_2)_2Li]^-\}_2[(hpp)_6Li_8]^{2+}$ **27**.[55]

Table 1 Bond lengths (Å) and angles (°) in the core of the empty cage $[(hpp)_6Li_8]^{2+}$ **27**$^{2+}$ at 180 K[55]

Li1–N3	1.987(5)	Li4–N8A	1.993(5)
Li1–N6	2.004(5)	C27–N1	1.365(3)
Li1–N8	1.973(5)	C27–N2	1.349(4)
Li2–N2	1.989(5)	C27–N3	1.349(4)
Li2–N6A	1.969(5)	C34–N4	1.353(3)
Li2–N7A	2.013(5)	C34–N5	1.359(4)
Li3–N3A	1.981(5)	C34–N6	1.340(4)
Li3–N4A	1.998(5)	C41–N7	1.348(3)
Li3–N7	1.964(5)	C41–N8	1.348(3)
Li4–N2A	1.981(5)	C41–N9	1.360(3)
Li4–N4	1.993(5)		
N2–C27–N3	117.6(2)	Li1–N6–Li2A	88.0(2)
N4–C34–N6	117.4(2)	Li3A–N4–Li4	88.2(2)
N7–C41–N8	117.0(2)	Li1–N8–Li4A	89.4(2)
Li1–N3–Li3A	89.4(2)	Li2A–N7–Li3	87.5(2)
Li2–N2–Li4A	89.5(2)		

of which we are aware being $[\{Et_2Al(\mu\text{-}OPh)_2\}_2Li]^-$ **30**$^-$, albeit the alkali metal now resides at the core of a tetraphenolate-defined tetrahedron.[59]

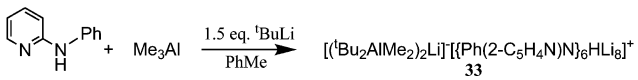
The counter-dication is based on an array of eight Li^+ ions in a nearly perfect cubic arrangement. Each face of this cube is straddled by an $[hpp]^-$ ion; conforming to the prerequisite that the amide anions must be able to anchor the otherwise mutually repelling alkali-metal centres. Each N-centre straddles two metal ions, the geometrically rigid ligand bite evidently ideal for a single guanidinate ion to bind the four metal ions of each cube face (Table 1). This high level of polyhedral regularity is consistent with the deployment of symmetrical guanidinate units and an analysis of bonding parameters reveals Li–N distances in the range 1.964(5)–2.013(5) Å (mean Li–N 1.987 Å) and mean $C(NLi_2)_2$ 117.3° in the $[hpp]^-$ ions. In the core, Li···Li non-bonding distances are in the range 2.751(7)–2.795(7) Å. These distances, it will be shown, are extended relative to those in analogous anion-encapsulation clusters and can, therefore, be viewed as being borne only of the conflicting effects of inter-metal electrostatic repulsion and the bonding action of the guanidinate ligands.

Occupation of the cage and proof of resident hydride

Our early work on the chemistry of mixed Li–Al systems quickly established that the treatment of amidoalanes with an organolithium substrate may result in much less straightforward chemistry than the simple nucleophilic generation of lithium aluminates.[47] Exposure of $AlMe_3$ to equimolar 2-*N*-pyridylaniline is known to yield the corresponding amidoalane $Ph(2\text{-}C_5H_4N)NAlMe_2$ **31** with the concomitant loss of methane.[60] However, to date the subsequent introduction of ^tBuLi has failed to give the aluminate $[Ph(2\text{-}C_5H_4N)NAlMe_2{}^tBu]Li$ **32**. Rather a single-crystalline material is reproducibly obtained. Whereas initial work utilized 1 eq. ^tBuLi it was soon noted that the employment of *excess* organolithium yielded better quality crystals. So far as has been observed, there is no discernable difference made to either the chemical identity of the isolated material nor to its yield if the amount of organolithium employed is raised from 1 to 1.5 eq. The use of more than 1.5 eq. ^tBuLi has invariably yielded either worse quality crystals (in depleted yields) or else has failed to give isolable product(s). Pending

Table 2 Selected bond lengths (Å) and angles (°) in the 1.5 K neutron diffraction structure of $[(^tBu_2AlMe_2)_2Li]^-[\{Ph(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ **33**[64]

Li5–C50X	2.23(2)	Li1–N3	2.05(2)
Li5–C51X	2.29(2)	Li1–N6A	2.10(2)
Al1–C50X	2.071(15)	Li2–N2A	2.14(3)
Al1–C51X	2.056(15)	Li2–N4	2.01(2)
Li5–H50A	2.26(2)	Li2–N5	2.01(2)
Li5–H50C	2.27(4)	Li3–N1	2.03(2)
Li5–H51A	2.21(4)	Li3–N3	2.05(2)
Li1–H	2.06(2)	Li3–N5	2.07(2)
Li2–H	1.92(2)	Li4–N1	2.04(2)
Li3 · · · H	2.85(2)	Li4–N4A	2.14(2)
Li4–H	2.04(2)	Li4–N6	2.05(2)
Li1–N2	2.01(2)		
C50X–Li5–C51X	98.1(3)	Li1–N3–Li3	83.7(8)
C50X–Al1–C51X	111.6(6)	Li2–N4–Li4A	71.9(9)
Li3–N1–Li4	85.8(9)	Li2–N5–Li3	82.5(9)
Li1–N2–Li2A	71.7(9)	Li1A–N6–Li4	74.7(8)



Scheme 2

crystallographic characterization it was revealed that, for the reaction done with 1.5 eq. ^{t}BuLi, the yield was 42%, based on that substrate. However, the beneficial effects on crystallinity mean that this procedure has been adopted consistently in this work. X-Ray crystallography first suggested the identity of this material to be $[(^tBu_2AlMe_2)_2Li]^-[\{Ph(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ **33** in 1999 (Scheme 2). As indicated in the previous section, the lithium bis(aluminate) anion components represent a recurring phenomenon in this field. Subsequent resolution of the neutron diffraction structure of $[(^tBu_2AlMe_2)_2Li]^-$ **33**$^-$ allowed a more exact interpretation of the bonding in this ion (Fig. 13(a), selected bonding parameters for the 1.5 K neutron diffraction structure are given in Table 2 and are discussed in detail in the text while selected data for the 180 K X-ray diffraction structure of **33** are given in Table 3). While species incorporating similar motifs to that noted in **33**$^-$ have been seen before,[57] the details of lithium ion stabilization have not been studied and, more generally, the area

Table 3 Selected bond lengths (Å) and angles (°) in the 180 K X-ray diffraction structure of the cationic μ_6-hydride component in **33**.[62] Atoms are labelled as for the 1.5 K neutron diffraction structure shown in Fig. 13(c)

Li1–H	2.037(7)	Li3–N3	2.063(9)
Li2–H	1.989(8)	Li3–N5	2.046(9)
Li3 · · · H	2.828(9)	Li4–N1	2.000(8)
Li4–H	2.019(7)	Li4–N4A	2.130(8)
Li1–N2	2.064(8)	Li4–N6	2.046(8)
Li1–N3	1.970(7)	C7–N1	1.356(5)
Li1–N6A	2.119(7)	C7–N2	1.373(5)
Li2–N2A	2.121(8)	C18–N3	1.364(6)
Li2–N4	2.044(9)	C18–N4	1.373(5)
Li2–N5	1.966(8)	C29–N5	1.367(6)
Li3–N1	2.008(9)	C29–N6	1.372(5)
N1–C7–N2	116.0(4)	Li1–N3–Li3	84.2(3)
N3–C18–N4	114.8(4)	Li2–N4–Li4A	73.4(3)
N5–C29–N6	115.8(3)	Li2–N5–Li3	84.3(4)
Li3–N1–Li4	84.9(4)	Li1A–N6–Li4	73.1(3)
Li1–N2–Li2A	72.9(3)		

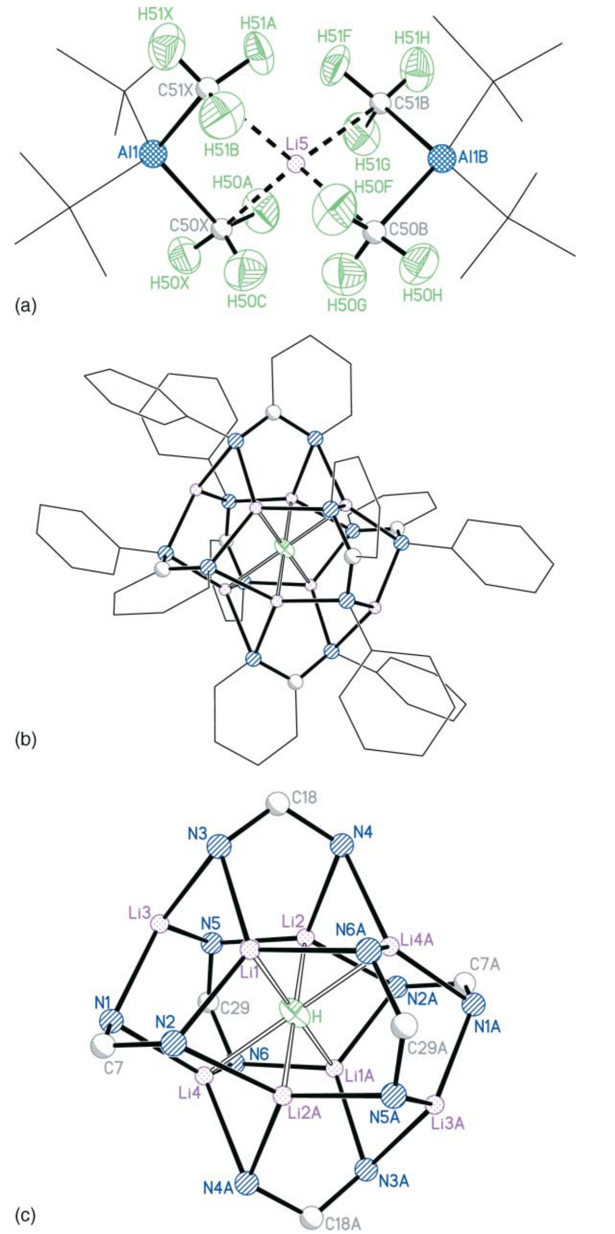
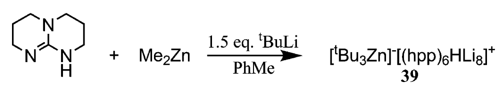


Fig. 13 (a) Neutron diffraction structure at 1.5 K of the trinuclear anion in $[(^tBu_2AlMe_2)_2Li]^-[\{Ph(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ **33**. Hydrogen atoms depicted at 50% probability.[64] (b) and (c) An initial 180 K X-ray diffraction analysis of the cation in $[(^tBu_2AlMe_2)_2Li]^-[\{Ph(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ **33** suggested the presence of octahedrally bound hydride[62] and this was subsequently proved by neutron diffraction.[64] The 1.5 K neutron structure of **33**$^+$ is shown, with the interstitial hydride represented at 50% probability.

of alkyl group interaction with alkali metals is poorly understood. In the present case, neutron diffraction fails to reveal clearly discernable C–H bond activation in the Al-bonded methyl groups. These distances lie in the range 1.01(2)–1.14(2) Å, an observation that supports the view that what is often described as being agostic stabilization in fact involves substantial C(Me) · · · Li interaction instead.[61] With neutron crystallography revealing mean values of C(Me)–Li 2.26 Å and C(Me)–Al 2.06 Å, the positions of the bridging carbon atoms with respect to either metal in **33**$^-$ correlate closely with those reported in polymeric **29** (mean C–Li 2.30, C–Al 2.02 Å).[57] Moving to the cation, whose charge must be +1 in order to achieve electrical neutrality, it follows that the cage architecture cannot be empty; that is, the 1 : 1 ratio of cations to anions in the crystal lattice precludes existence of the octa-lithium hexaamide dication. Detailed analysis of the Fourier difference map, generated from the otherwise complete X-ray model, had already revealed a concentration of residual electron density in the centre of the alkali-metal cage at a level attributable to interstitial hydride.[62] That said, the failure of ^{1}H NMR spectroscopy to locate a definitive resonance for the hydride ion made proving the identity of the interstitial anion a priority. With that in mind, the new Very-Intense Vertical-Axis Laue Diffractometer (VIVALDI)[63] at the Institut Laue-Langevin was employed to carry out a single-crystal neutron diffraction study on the material. This instrument employs Laue diffraction using a polychromatic thermal-neutron beam with a large solid-angle cylindrical image-plate detector to raise the detected diffracted intensity by one-to-two orders of magnitude, compared with a conventional monochromatic experiment. The result was that a successful data collection was possible on a 2 × 2.5 × 0.5 mm crystal with a unit cell volume of 8234(3) Å^3, with experiments undertaken at 180 and 1.5 K.[64] However, significant thermal motion at the higher temperature dampened the data intensity such that, whilst the structure could be solved, data only allowed anisotropic refinement of the interstitial ion. At 1.5 K, the maximum diffraction intensity could be obtained, minimizing extraneous nuclear density in the (interstitial) site of interest and the interstitial hydride ion could be unambiguously and entirely located in the Fourier difference map. The cationic cluster was found to comprise a $(Li^+)_8$ cubic cage whose six faces are straddled by the backbones of the six pyridylamide ligands in a fashion akin to that noted above in the unoccupied **27**$^{2+}$ cage. However, significant perturbation renders the $(Li^+)_8$ cluster a distorted bicapped octahedron (Fig. 13(b), Tables 2 and 3, Chart 2). Li3 and its symmetry equivalent represent the two *trans* caps, lying essentially symmetrically over the Li1/2/4 triangular face and its symmetry equivalent (mean non-bonding Li3 · · · Li 2.73 Å by neutron diffraction at 1.5 K). The non-bonding Li · · · Li distances in the central octahedron exhibit considerable variations and lie in the range 2.43(3)–3.23(3) Å. Importantly, these perturbations correlate with variations in N–Li bonds that derive from asymmetry in the 2-*N*-pyridylanilide ligands that were lacking in the symmetrical guanidinate [hpp]$^-$ ions. The six metal ions in the core bond to just one formally deprotonated amide and two neutral pyridyl N-centres, whilst the two extruded ions bond to three formally deprotonated N-centres (N1, N3 and N5 for Li3 in Fig. 13(c)). In spite of these variations, metal-nitrogen distances in the neutron structure are essentially uniform; mean N–Li3 2.05 Å, mean N–Li1/2/4 2.06 Å. While metal-nitrogen distances appear unaffected, hydride coordination is strongly dependent upon the positioning of the Li$^+$ ions and, therefore, of the charged amide centres. The unique nitrogen bonding modes for the extruded metals, and the resultant extension of Li · · · Li distances involving them, yield an octahedrally coordinated hydride ion (H–Li1 2.06(2), H–Li2 1.92(2), H–Li4 2.04(2) Å). The extruded ions plainly fail to interact with the interstitial hydride (H · · · Li3 2.85(2) Å) and, overall, it is possible to suggest a clear correlation between the properties of the N-donor centres (neutral or charged) and whether or not the alkali metals with which they interact are hydride-coordinated. Lastly, the octahedral hydride ion pleasingly resembles that known in the neutron diffraction structure of $(LiH)_\infty$ **34** (Li–H 2.04 Å).[65]

Choosing reagents, 1: geometrical variations in the cage

The observation of an octahedral bonding pattern in the cationic component of **33** by no means suggests that hydride coordination modes in such systems will be limited to μ_6. That said, trivial alterations to reaction conditions have indicated a preference for six-fold hydride coordination; $[Me(Al^tBu_3)_2]^-[\{(2\text{-}MeC_6H_4)(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ **35** was obtained using 2-*N*-pyridyl(2-methylaniline),[15] $[(^tBu_2AlMe_2)_2Li]^-[\{(4\text{-}MeC_6H_4)(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ **36** resulted from the use of 2-*N*-pyridyl(4-methylaniline),[66] and the employment of $AlEt_3$ gave $[(^tBu_2AlEt_2)_2Li]^-[\{Ph(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ **37**.[15] In all of these, the cluster cation is precisely analogous to that described in detail for **33**$^+$. The ability of a *more symmetrical* ligand to induce the creation of a more geometrically regular cubic cage has already been discussed in the context of **27**, and we therefore sought to generate hydride encapsulation clusters incorporating the symmetrical [hpp]$^-$ ligand. Our inability to synthesize (or, at least, to isolate) $[(^tBu_2AlMe_2)_2Li]^-[(hpp)_6HLi_8]^+$ **38** led us to experiment with the use of other Lewis acids and, based on other work going on in the group,[67] Me_2Zn was selected as a suitable candidate. The ligand hppH was treated sequentially with Me_2Zn and ^tBuLi in toluene, forming an oil from which crystals were obtained. ^{1}H NMR spectroscopy indicated a 2 : 1 ratio of hpp to ^tBu units and, pleasingly, X-ray crystallography revealed ion-separated $[^tBu_3Zn]^-[(hpp)_6HLi_8]^+$ **39** (35% based on ^tBuLi, Fig. 14, Scheme 3, Table 4, Chart 3, the lattice contained half a toluene molecule per hydride). Two crystallographically independent, separated



Scheme 3

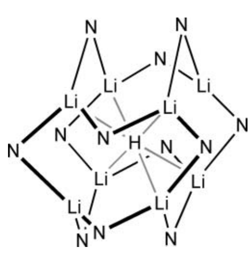


Chart 2

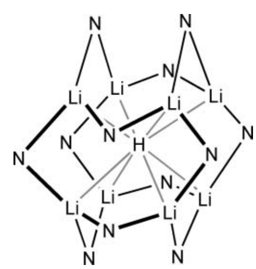


Chart 3

Table 4 Selected bond lengths (Å) and angles (°) in μ_8-hydride $[^tBu_3Zn]^-[(hpp)_6HLi_8]^+$ **39** at 173 K[55]

Li1–H	2.21(1)	Li4–N2	1.974(11)
Li2–H	2.17(1)	Li4–N5A	2.031(11)
Li3–H	2.16(1)	Li4–N7	2.001(11)
Li4–H	2.10(1)	C13–N1	1.349(8)
Li1–N1	2.019(11)	C13–N2	1.316(8)
Li1–N4	2.014(11)	C13–N3	1.396(8)
Li1–N7	2.029(10)	C20–N4	1.353(7)
Li2–N1A	2.026(11)	C20–N5	1.329(7)
Li2–N5A	1.984(11)	C20–N6	1.386(7)
Li2–N8	1.999(11)	C27–N7	1.340(7)
Li3–N2	2.007(11)	C27–N8	1.332(7)
Li3–N4A	2.000(11)	C27–N9	1.377(7)
Li3–N8A	2.030(10)		
N1–C13–N2	119.1(5)	Li1–N4–Li3A	78.0(4)
N4–C20–N5	118.9(5)	Li2A–N5–Li4A	76.0(4)
N7–C27–N8	117.6(5)	Li1–N7–Li4	76.2(4)
Li1–N1–Li2A	76.8(4)	Li2–N8–Li3A	76.5(4)
Li3–N2–Li4	77.4(4)		

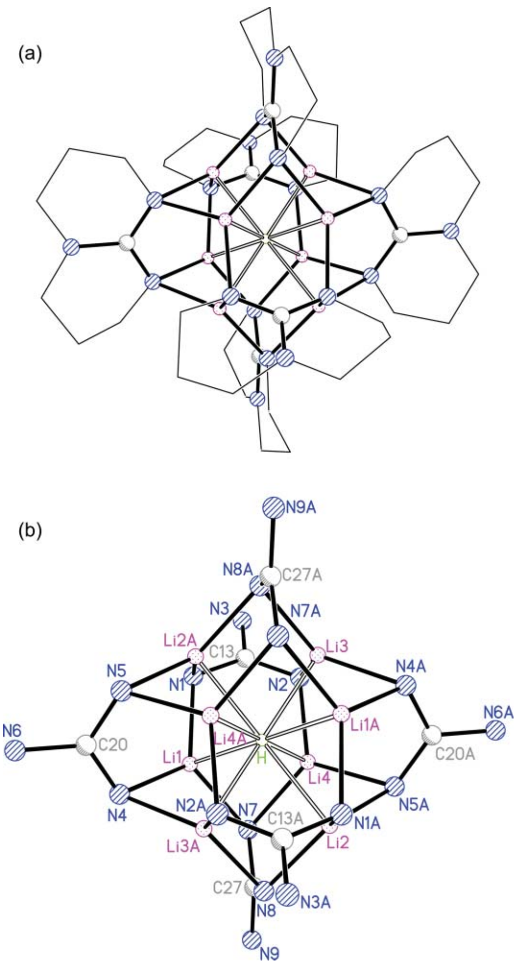


Fig. 14 In contrast to the empty cluster noted for hydride-free aluminate **27**, the use of an organozinc substrate allows the isolation and analysis of the μ_8-hydride $[^tBu_3Zn]^-[(hpp)_6HLi_8]^+$ **39**.[55]

ion-pairs exist in the unit cell and one representative set is discussed here. As was the case with aluminium, formation of the anionic component was non-trivial and, in this instance, a near planar, anionic tris(*tert*-butyl)zincate ion is observed. The counter-cation incorporates an almost exactly cubic array of closely packed alkali-metal centres synonymous with use of the symmetrical $[hpp]^-$ ligand and, as seen previously, these residues act to cap each face of the core. Contrary to what was seen in the refinement of hydride-free $\mathbf{27}^{2+}$, interstitial electron density was now observable at the centre of the cluster, and close analysis of the Fourier difference map allowed the satisfactory refinement of an interstitial hydride. Taken together with the proof of principle provided by the neutron diffraction analysis of $\mathbf{33}^+$ the case for interstitial hydride is compelling. However, the coordination mode reported in the neutron structure is not replicated here. Instead, the essential regularity of the alkali-metal cluster $\mathbf{39}^+$ results in the observation of four near-equal metal–hydride distances (mean 2.16 Å, Li1–H 2.21(1), Li2–H 2.17(1), Li3–H 2.16(1), Li4–H 2.10(1) Å). These values are greater than those seen in **33** and **34** and are consistent with the interstitial ion now being μ_8-coordinated.[64,65]

We now had the first evidence that manipulating cage geometry, through the selection of appropriate ligands, could induce changes to the level of hydride coordination. The inclusion of hydride has a second interesting effect on the $[(hpp)_6Li_8]^{2+}$ superstructure. As pointed out above, the significant extrusion of two metal ions that has been noted when employing asymmetric pyridylaniline substrates is not reproduced here. It is clear, however, that in $\mathbf{39}^+$ Li1 and its symmetry equivalent are nominally extruded. They therefore participate in the longest of the Li–H bonds (2.21(1) Å) and in the longest cube-edge distances (mean Li1 · · · Li2A/3A/4 2.509 Å). The dimensions of the cube are of particular importance because they give us a means of measuring the effect upon the inter-metal displacements of hydride encapsulation. In spite of employment of the same organic ligand and the presence of the same number of metal ions, the inter-metal cube-edges if interstitial hydride is incorporated prove to be significantly reduced relative to those in the empty $\mathbf{27}^{2+}$ cage. Thus, it is possible to compare a Li · · · Li range of 2.473(14)–2.527(14) Å (mean 2.497 Å) in $\mathbf{39}^+$ with one of 2.751(7)–2.795(7) Å (mean 2.777 Å) in $\mathbf{27}^{2+}$. This flexibility in the cluster core vexes the question of whether the lithium–nitrogen bonds modulate in response to the different electrostatic requirements of the metals in either cluster? This indeed appears to be the case. The Li–N distances in $\mathbf{27}^{2+}$ (mean 1.987 Å, range 1.964(5)–2.013(5) Å) are shorter than those in $\mathbf{39}^+$ (mean 2.010 Å, range 1.984(11)–2.030(10) Å). These differences in Li · · · Li and Li–N distances can be confidently attributed to *only* whether the hydride is present or not. In its absence, inter-metal repulsion dominates and the metal array expands. The bite angle of the ligand changes only marginally—a mean $C(NLi_2)_2$ angle of 118.6° in $\mathbf{39}^+$ (range 119.1(5)–117.6(5)°) compares with a mean $C(NLi_2)_2$ angle of 117.3° in $\mathbf{27}^{2+}$ (range 117.6(2)–117.0(2)°)—but bonds to the organic anions clearly contract in response to expansion of the metal array while the Li–N–Li angle also necessarily expands in the absence of interstitial hydride. Mean variations in $\mathbf{27}^{2+}$ and $\mathbf{39}^+$ are summarized in Fig. 15.

The most prevalent anion encapsulation behaviour noted for lithium amide systems is oxide incorporation. While the field is dominated by μ_6- and μ_8-encapsulation, it is also apparent that μ_2-bonding of the interstitial ion is possible. In the context of ion encapsulation this has taken the form of the incorporation (within a higher lithium-based aggregate) of a linear Li_2O molecule[13–15] (although bent configurations are known, in these the oxide is no longer interstitial).[60] It was therefore decided to investigate

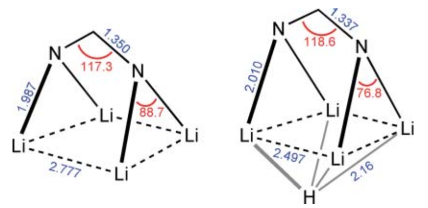
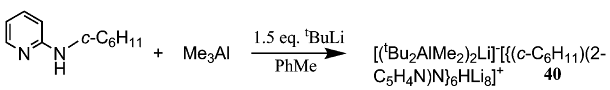


Fig. 15 Comparison of mean interatomic distances (Å) and angles (°) in the cores of 27^{2+} (180 K, left) and 39^{+} (173 K, right).[55]

whether the same kind of pattern of low-coordinate bonding could pertain for hydride encapsulation chemistry. In order to do this the electronic properties of the N-donor ligands were evaluated. We asserted that in the structures noted thus far, the extrusion of metal centres from the cluster core was synonymous with charge donation from the ligand (that is, better donation from the ligand obviated the need to accept charge from interstitial hydride). Accordingly, interaction with formally charged N-centres in $\mathbf{33}^{+}$ and $\mathbf{35}^{+}$-$\mathbf{37}^{+}$ incurred the removal of two metal ions from the core, whereas the remainder of the Li^{+} ions were at least partially stabilized by the neutral pyridyl group. Moreover, the use of a symmetrical ligand in which charge was evenly distributed removed all but the most minor extrusion and left the hydride eight-coordinate. This led us to test the effect of employing $HN(2\text{-}C_5H_4N)R$, in which R was an electrorepellent alkyl group. The chosen ligand, in which R = $c\text{-}C_6H_{11}$, was reacted with $Me_3Al/^tBuLi$ and the crystalline deposit formed was found to be $[(^tBu_2AlMe_2)_2Li]^-[\{(c\text{-}C_6H_{11})(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ **40** (43%, based on ^{t}BuLi, Fig. 16, Table 5, Scheme 4, Chart 4).[66] The anion is precisely analogous to that noted on several occasions previously. The cation though, which again revealed directly observable interstitial electron density in the Fourier difference map, showed subtle differences to those already described. The located hydride is now best viewed as being linearly μ_2-coordinated (Li3–H 1.839(10) Å). This value is notably shorter than even the shortest bonding distances in the μ_6-cages.[15,62,64] All of the remaining Li^{+} ions are displaced from the cluster core and, though the expansion is less pronounced than the extrusion noted in the $\mathbf{33}^{+}$ ion (Li1 ··· H 2.396(13) Å, Li2 ··· H 2.351(14) Å, Li4 ··· H 2.378(13) Å, mean 2.375 Å), the resulting (non-bonding) distances are considerably longer than those in the core, linear $[Li_2H]^+$ fragment. As was the case with $\mathbf{33}^{+}$, the imperative for extrusion is closely related to ligand structure and derives from charge donation by the



Scheme 4

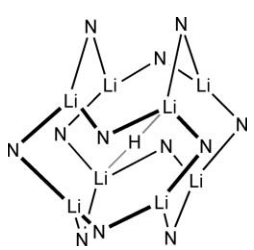


Chart 4

Table 5 Bond lengths (Å) and angles (°) in the core of μ_2-hydride $[\{(c\text{-}C_6H_{11})(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ $\mathbf{40}^{+}$ at 180 K[66]

H ··· Li1	2.396(13)	N3A–Li3	2.022(13)
H ··· Li2	2.351(14)	N5–Li3	2.024(13)
H–Li3	1.839(10)	N2–Li4	2.060(13)
H ··· Li4	2.378(13)	N4–Li4	2.053(13)
N2A–Li1	2.060(13)	N5A–Li4	2.113(13)
N3–Li1	2.137(13)	C5–N1	1.381(9)
N6–Li1	2.065(13)	C5–N2	1.374(9)
N1–Li2	2.139(13)	C16–N3	1.396(9)
N4–Li2	2.105(13)	C16–N4	1.352(9)
N6–Li2	2.050(14)	C27–N5	1.386(9)
N1–Li3	2.033(12)	C27–N6	1.357(9)
N1–C5–N2	115.4(8)	Li1-N3-Li3A	77.1(5)
N3–C16–N4	116.2(8)	Li2–N4–Li4	78.6(5)
N5–C27–N6	117.3(7)	Li3–N5–Li4A	76.4(5)
Li2–N1–Li3	77.3(5)	Li1–N6–Li2	79.2(5)
Li1A–N2–Li4	78.0(5)		

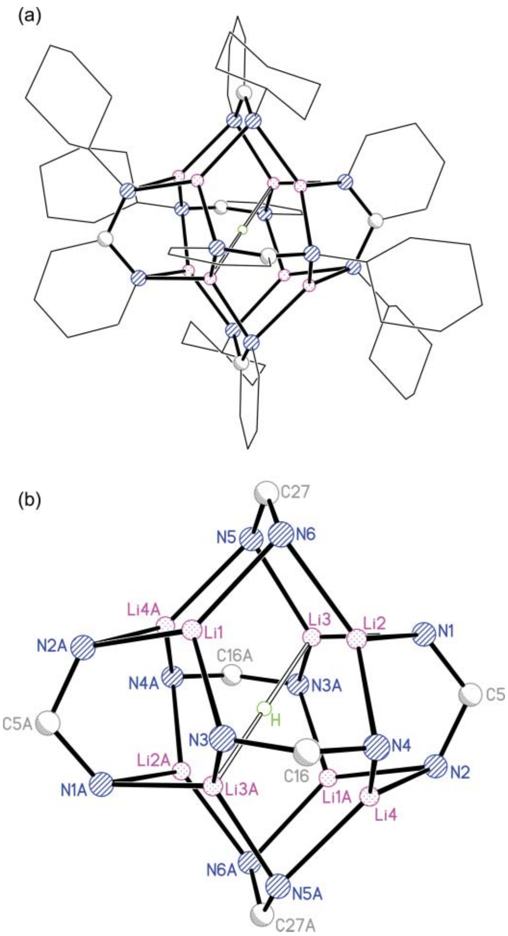


Fig. 16 Manipulation of hydride coordination has proved possible by incorporating electrorepellent cycloalkyl moieties in the cationic component of $[(^tBu_2AlMe_2)_2Li]^-[\{(c\text{-}C_6H_{11})(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ **40**.[66]

amide. Hence, extruded ions Li1/2/4 and their symmetry equivalents interact with only a single neutral pyridyl ring and with two formally deprotonated amide N-centres. However, Li3 and Li3A interact solely with three neutral pyridyl N-atoms (N1, N3A and N5 for Li3 in Fig. 16) and so accept charge from the interstitial hydride as well. The 'intermediacy' of the Li1/2/4 ··· H distances between the plainly bonding and non-bonding distances reported in the μ_6-cages vexes the question; at which point does metal-hydride interaction cease? Nevertheless, it is clear that the extreme shortness of the Li3–H length renders the structure of **40**$^+$ unique. Mean interatomic parameters in the 180 K X-ray diffraction structures of **33**$^+$ and **40**$^+$ are compared in Fig. 17, with the cyclohexyl system failing to show the significant distortions (*cf.* variations in Li · · · Li and Li–N–Li) noted in the pyridylamide analogue.

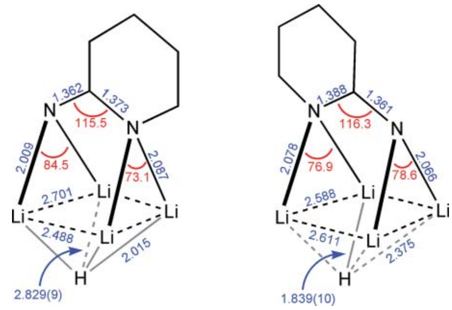


Fig. 17 Comparison of mean interatomic distances (Å) and angles (°) in the cores of the 180 K X-ray diffraction structures of **33**$^+$ (left) and **40**$^+$ (right).[62,66]

Choosing reagents, 2: extending the choice of Lewis acid

So far, the importance of a Lewis-acid component in achieving this chemistry has been demonstrated by employing organoaluminium or organozinc reagents. More recently, boranes were chosen as potentially useful Lewis acids with which to generate and trap hydride. To our surprise, simple triethylborane proved to be unreactive towards amines. This contrasted with our observations on the results of treating N-donor ligands with either Al or Zn reagents, in which cases amidoalane **31** and $\{R(2\text{-}C_5H_4N)N\}_nZnMe_{2-n}$ (R = Ph, n = 2 **41**; R = c-C_6H_{11}, n = 1 **42**) could be isolated, respectively (Chart 5).[60,68]

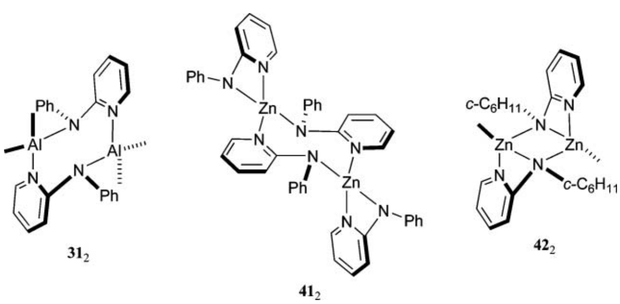


Chart 5

In spite of the evident passivity of boranes with respect to amines, initial attempts to combine Et_3B with 2-N-pyridylaniline and ^{t}BuLi were promising. However, X-ray diffraction proved frustrating; whilst the formation of a distorted **33**$^+$-type cation was unambiguous (interstitial electron density could even be refined), the counter-anion could neither be confidently resolved crystallographically nor extrapolated from other analytical data. In a similar vein to that noted above, whereas guanidinidoalane **28** and $Me_2(hpp)_4Zn_3$ **43** (Chart 6) have been documented,[56,68]

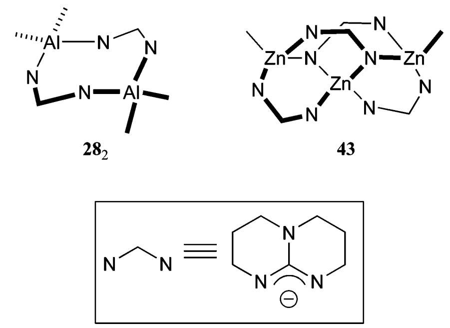
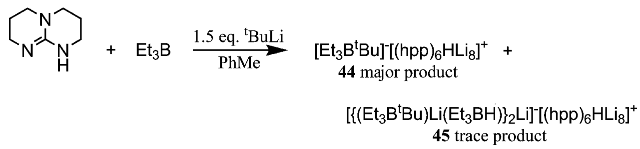


Chart 6

an Et_3B-hppH mixture fails to afford a detectable reaction and NMR spectroscopy suggests that the only material to redeposit is unreacted guanidine. In spite of this, tentative data obtained using 2-N-pyridylaniline led to the expectation that the introduction of ^{t}BuLi should still afford hydride capture. Multiple single-crystal X-ray diffraction experiments and analysis of the bulk product by spectroscopic methods has borne out this view, with crystalline material isolable from this reaction being identified as the hydride-capture complex $[Et_3B^tBu]^-[(hpp)_6HLi_8]^+$ **44** (27%, based on guanidine, Scheme 5). In accord with the previously noted **39**$^+$ superstructure, each face of a near cubic array of metal ions is capped by $[hpp]^-$ and the Fourier difference map reveals observable hydridic interstitial electron density that is μ_8-coordinated (mean Li–H 2.16 Å).[66] There is no question that **44** is the dominant isolable product. However, some evidence does exist for the presence of other products, and a, presumably rogue, crystal structure of the remarkable species $[\{(Et_3B^tBu)Li(Et_3BH)\}_2Li]^-[(hpp)_6HLi_8]^+$ **45** has been obtained (Fig. 18, Scheme 5).[69] As we shall see below, the formation of borohydrides in systems such as these proves highly instructive. The cationic component shows the (by now) predictable μ_8-coordinated hydride. However, the oligomeric counter-anion merits a particular mention by virtue of its implications for



Scheme 5

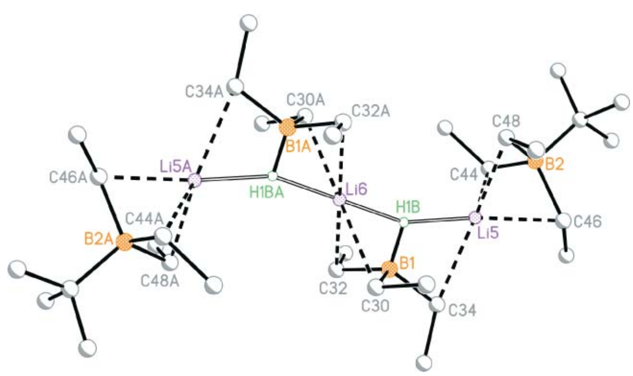


Fig. 18 The oligomeric anion in the trace species $[\{(Et_3B^tBu)\text{-}Li(Et_3BH)\}_2Li]^-[(hpp)_6HLi_8]^+$ **45**.[15]

the mechanism of hydride generation in systems such as these. It comprises B(μ-C)Li and B(μ-H)Li motifs and terminal and bridging borate ions that render Li5 distorted tetrahedral and Li6 distorted square planar. Direct observation of μ_3-hydride H1B in the $[Et_3BH]^-$ ion is allowed by Fourier difference synthesis (H1B–B1 1.23(3), H1B–Li5 1.86(3), H1B–Li6 1.83(3) Å).

It was this observation of $[Et_3BH]^-$ units that first lead us to postulate, albeit tenuously, a role for *in situ* generated simple molecular main-group hydrides as precursors to the formation of group 1 interstitial hydride clusters. Subsequent supporting evidence amassed took the form of the crystalline deposit obtained in the wake of attempts to generate hydride encapsulation clusters in the presence of tmeda. This material analysed by NMR spectroscopy as $[Et_3BH]^-[Li{\cdot}2tmeda]^+$ **46**.[66] Unlike in our previous work, utilizing Al or Zn Lewis acids, the presence of boron presents us with a convenient multinuclear NMR opportunity and has allowed definitive identification of the hydride through the observation of a doublet at δ −13.09 ($^1J_{BH}$ 74.6 Hz). The simple ion-separate nature of **46** was corroborated by a preliminary crystallographic study (Chart 7).[15]

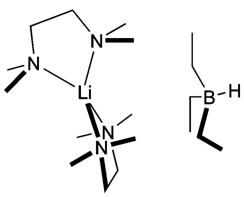


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Chart 7

Based on the idea that the formation of $[Et_3BH]^-$ is central to hydride encapsulation (see calculations below) it was considered that in the presence of an *N*,*N*′-bidentate lithium amide any supplier of LiH should be capable of encouraging the formation of analogous dicationic superstructures capable of capturing hydride. This theory was tested by treating the pre-formed guanidinate hppLi[16] **47** with Et_3BHLi in the ratio 3 : 1 (designed to supply the components necessary to form $[Et_3BH]^-[(hpp)_6HLi_8]^+$ **48**). In the event, this protocol gave ion-separate $[(Et_3B)_2H]^-[(hpp)_6HLi_8]^+$ **49** (21%, with the inclusion of an unreacted $[Et_3BH]^-$ unit in the anion making borohydride the limiting reagent, Fig. 19(a), Scheme 6). The hydride encapsulating cation is precisely analogous to that already discussed at length for **39**$^+$. The main point is that these data suggest hydride encapsulation proceeds with 3 : 1 stoichiometry and with 2 eq. Et_3BHLi contributing $H^- + 2Li^+$ to the formation of each cluster. That, of course, requires the concomitant formation of 1 eq. of each of Et_3B and $[Et_3BH]^-$, with subsequent adduct formation logically explaining the counter-anion. This anion is, in fact, extremely unusual. Just a single example could be located of the crystallographic characterization of an unsupported monohydride bridge between boron atoms. Moreover, in contrast to the linear structure noted for **49**$^-$, the $[B_2H_7]^-$ ion showed a significant 127(2)° bend.[70] However, vibrational spectroscopy studies[71] and our own more recent DFT

$$\underset{\mathbf{47}}{6\ \mathrm{hppLi}} + 2\ \mathrm{Et_3BHLi} \xrightarrow{\mathrm{PhMe}} \underset{\mathbf{49}}{[(\mathrm{Et_3B})_2\mathrm{H}]^-[(\mathrm{hpp})_6\mathrm{HLi_8}]^+}$$

Scheme 6

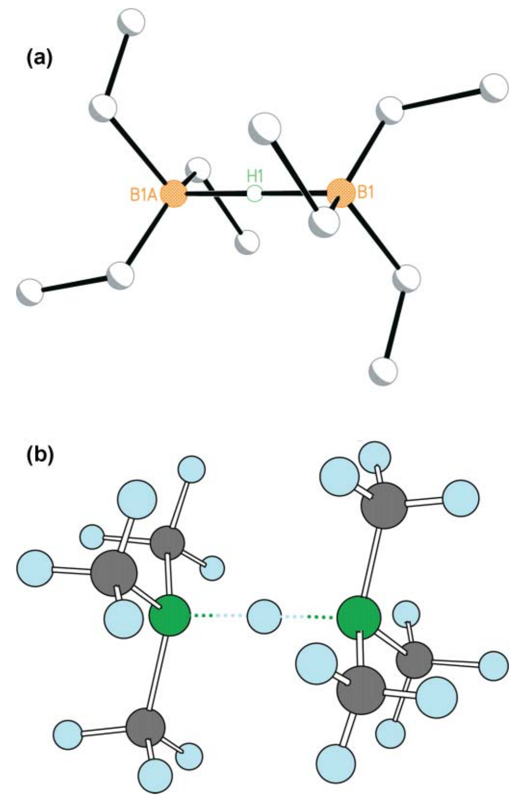


Fig. 19 (a) The near linear anionic Lewis adduct component of $[(Et_3B)_2H]^-[(hpp)_6HLi_8]^+$ **49**.[66] (b) DFT optimisation of the model ion $[(Me_3B)_2H]^-$ **50**$^-$ shows a preference for a 179.4° B–H–B angle. The illustrated structure was 6.3 kcal mol^{-1} more stable than one modelled with a frozen 127.0° angle.[66]

calculations confirm a preference for linear bridging; it has been found that the model borate $[(Me_3B)_2H]^-$ **50**$^-$ prefers an essentially linear structure (calculated B–H–B 179.4°) to a frozen 127° bend by 6.3 kcal mol^{-1} (Fig. 19(b)). Unfortunately, in spite of the convenience with which the reaction apparently balances, the system is evidently more complex, and both ^{1}H and ^{11}B NMR spectroscopy suggest the trace presence of a second component to the crystalline deposit. Accordingly, the ^{11}B NMR spectrum of crystalline **49** reveals a dominant doublet at δ −13.07 ($^1J_{BH}$ 73.3 Hz). The minor impurity is observed at δ −16.64 and can be attributed to the presence of lithium tetraethylborate **51**-confirmed by the synthesis of a control sample from EtLi and Et_3B. In spite of the competitive formation of (at least trace) **51**, it is clear that the direct combination of a lithium hydride source with a preformed *N*,*N*′-bidentate lithium amide can indeed yield hydride encapsulation merely through a translocation process.

Choosing reagents, 3: the role of ligand structure

The effects of utilizing both asymmetric and symmetric *N*,*N*′-bidentate ligands on the issue of symmetry in the pseudocubic $[\{R(2\text{-}C_5H_4N)N\}_6Li_8]^{2+}$ (R = aryl, cyclohexyl) and $[(hpp)_6Li_8]^{2+}$ superstructures have already been noted. Moreover, it has proved possible to correlate distortions in these frameworks with the (de)localization of charge density in the ligand shells. Hence, asymmetric $[N(2\text{-}C_5H_4N)R]^-$ ligands induce the compression and extrusion of alkali-metal centres and afford a route to the manipulation of hydride coordination.[62,66] Contrastingly, the highly symmetrical guanidinate [hpp]$^-$ yields near perfect pseudocubic symmetry either in the presence of or absence of an interstitial ion.[55] Logically, the successful employment of guanidinate ligands suggests the viability of amidinates in this field. These have hitherto been employed in the study of group 13 hydrides,[72] and attempts to replicate hydride encapsulation clusters using a variety of amine, guanidine and amidine substrates are ongoing. However, in the same way that steric shielding by [DippNC(H)NDipp]$^-$ (Dipp = diisopropylphenyl) has induced kinetic control in indium hydride chemistry by reducing the accessibility of the hydride and also by circumventing the formation of a M–H–M motif,[73] some interesting steric effects have been seen to operate when attempting to utilize *N*,*N*′-diphenylbenzamidine in conjunction with Me_3Al/^tBuLi. Whereas the expected result, ion-separate $[(^tBu_2AlMe_2)_2Li]^-[\{PhNC(Ph)NPh\}_6HLi_8]^+$ **52**, should reveal integrals for the respective organic components of 90 : 12 : 36, a 45 : 18 : 18 pattern is observed, which most obviously contains half the required amount of amidinate and equivalent integrals attributable to ^{t}Bu and Me groups. This is consistent with a hydride-free cluster of the type $\{[(^tBuAlMe_3)_2Li]^-\}_2[\{PhNC(Ph)NPh\}_6Li_8]^{2+}$ **53** in which the bis(aluminate) anion has a slightly different organic composition to that noted before (*viz.* **27**).[55] However, X-ray diffraction revealed a different explanation, the structure of $[(^tBuAlMe_3)_2Li]^-[Am_3Li_4]^+$ (Am = PhNC(Ph)NPh) **54** (Fig. 20, Scheme 7, Chart 8).[74] The anionic component does indeed revisit the lithium bis(aluminate) motif that has already been described, notwithstanding the 3 : 1 Me : ^{t}Bu ratio. This ratio appears to be central to the ability of ion-association to now occur. Hence, **54**$^-$ contains two Al centres arranged such that two methyl substituents on each are disposed tetrahedrally about the central lithium ion. However, the incorporation of a methyl group terminally bonded to the electropositive Al centre gives an electron-rich ligand capable of interacting with the counter-ion and inducing polymerization (Li2 ⋯ C14 and Li4A ⋯ C5 give a mean Li ⋯ C of 2.295 Å).[75]

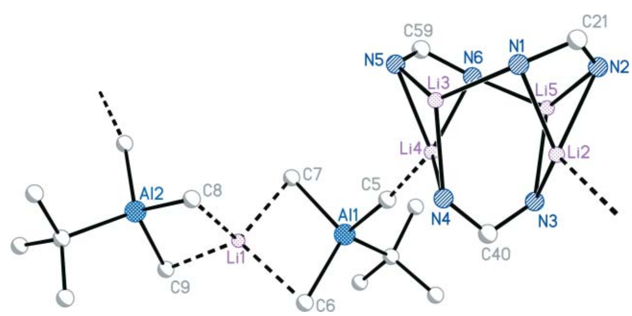
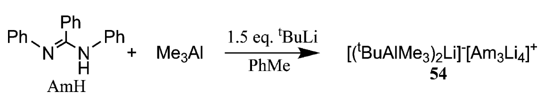
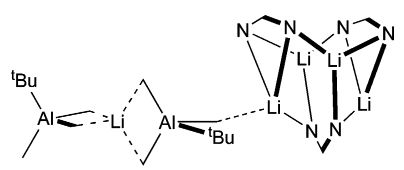


Fig. 20 X-Ray crystal structure of the chain that results from weak ion-association in $[(^tBuAlMe_3)_2Li]^-[\{PhNC(Ph)NPh\}_3Li_4]^+$ **54** (Ph rings omitted).[74]



Scheme 7

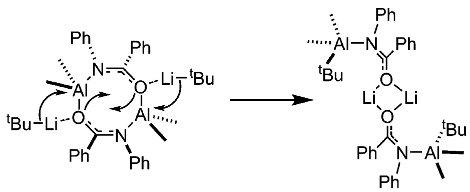


54

Chart 8

Like the $[Am_6Li_8]^{2+}$ (Am = amide, guanidinate) superstructures reported here, the **54**$^+$ ion both takes the form of a superlithiated cluster and reveals a Li : organic ratio of 4 : 3. It appears that it is the steric bulk of the *N*,*N*′-diphenylbenzamidinate ligand that inhibits the size of the cation aggregate. This incorporates two types of amidinate ligand, each of which interact with metal centres using both of their donor atoms. The two that incorporate N1⋯C21⋯N2 and N5⋯C59⋯N6 straddle Li3 and Li5 to give a concave eight-membered $(NCNLi)_2$ ring.[76–78] The Li–N interactions within this ring are short (1.992(6)–2.016(6) Å) in comparison to those between these nitrogen atoms and the remaining two metal ions in the cation (Li2/4, range 2.064(7)–2.199(6) Å). These interactions result from the bidentate chelation of Li2/4—a bonding mode not seen in superlithiated pseudocubic $[Am_6Li_8]^{2+}$ ions, in which the ligands straddle metal centres rather than chelating them. Chelation, however, is asymmetric (mean Li2–N1, Li4–N6 2.077 Å, mean Li2–N2 and Li4–N5 2.195 Å), a pattern consistent with the bonding in the eight-membered $(NCNLi)_2$ ring. Hence, N1/6 form relatively short bonds to Li2/4 but interact more weakly with Li- (mean Li–N 2.016 Å) and C-centres (mean N⋯C 1.342 Å) within the ring, while N2/5 form stronger bonds with Li- (mean Li–N 1.994 Å) and C-centres (mean N⋯C 1.325 Å) in the ring at the expense of short interactions with the two basal metals. These last two metal centres are straddled by the third ligand, and this is orientated such that neither Li2 ⋯ N4 nor Li4 ⋯ N3 (mean 2.583 Å) can be regarded as bonding interactions. Thus, this part of the cluster takes the form of an $(LiN)_2$ ring that has been fractured to give two LiN fragments,[79] presumably by virtue of the coordination of these Li^+ centres by bis(aluminate) anions.

The effect of varying the identities of ligand donor centres has proved to be dramatic. In seeking to extend the proven ability of Lewis acids of both Al and Zn to induce hydride production, it was decided to test both Lewis-acid types in conjunction with potentially *N*,*O*-bidentate carboxylic amides. In contrast to hydride encapsulation behaviour, however, the formation of alkali-metal imidates is observed. Reaction of Me_3Al with *N*-phenylbenzamide PhC(O)N(Ph)H forms a dimer of the amidoalane $PhC(O)N(Ph)AlMe_2$ **55** that is based on an eight-membered $(OCNAl)_2$ core.[80] In contrast to the behaviour of other amidoalane reagents, however, this dimer reacts with ^{t}BuLi to give a crystalline product that NMR spectroscopy suggests to be the simple aluminate $[PhC(O)N(Ph)Al(Me)_2{}^tBu]Li$ **56**.[81] X-Ray diffraction reveals a dimeric formulation that can be rationalised by arguing that the dimeric **55**$_2$ precursor reacts *via* a complex induced proximity effect (Scheme 8). The resulting aluminate dimer incorporates an essentially planar $(LiO)_2$ ring resulting from the formation of a short Li1–O2 bond (1.846(9) Å) and a longer Li2–O1 interaction (1.911(8) Å) with intramonomer stabilization of the Li^+ ions supplied by Li1–O1 (1.896(8) Å) and Li2–O2 (1.924(9) Å) interactions. In tandem with the donor behaviour of oxygen rather than nitrogen, evaluation of the backbones in the organic ligands (N1–C13 1.322(5), N2–C32 1.291(6),



($55 \cdot {}^tBuLi)_2$ 56_2

Scheme 8

O1–C13 1.270(5), O2–C32 1.291(6) Å) suggests the formation of lithium imidate units.[82,83] *Cis*, rather than (normally preferred) *trans*,[84] isomerism is revealed about the N · · · C bond (Fig. 21(a)). Peripheral to the ring core, the alkali metals are stabilized through the formation of weak alkyl interactions comparable in length to those in previously reported lithium 'ate complexes in which the group 1 centres were otherwise two-coordinate.[85,86] The different coordinative modes adopted by each lithium with respect to these alkyl groups (Li1 is stabilized by bonds with both Me and ^{t}Bu groups, whereas Li2 bonds to only Me) ties in with the formation of an interaction with an adjacent dimer in the crystal lattice. The association of four dimers gives an octanuclear array (Fig. 21(b)).

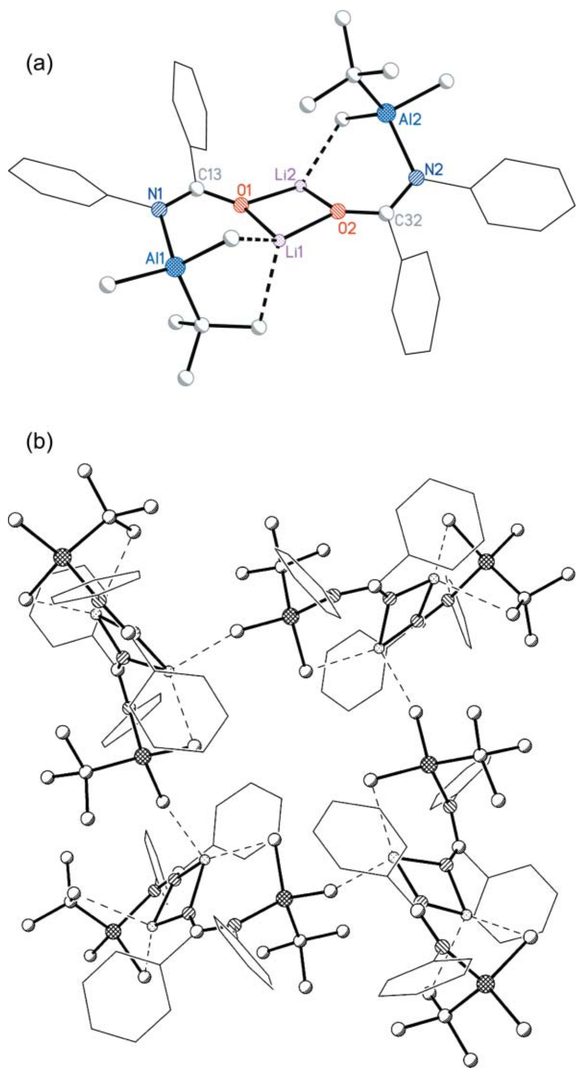
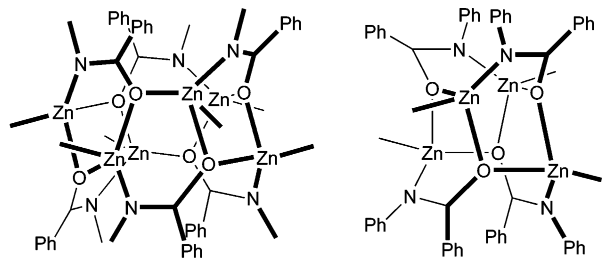


Fig. 21 (a) Molecular structure of the dimer of [PhC(O)N(Ph)-Al(Me)$_2$^{t}Bu]Li **56**.[81] (b) The octanuclear arrangement of dimers of **56**.[81]

In a similar vein to that noted for aluminium chemistry, the employment of organozinc Lewis acids in conjunction with carboxylic amides and a lithiating reagent has afforded simple lithium zincates. Less was known about the zinc amide precursors to zincate formation[87] and, it transpired, the reaction of *N*-methyl benzamide with Me_2Zn in non-donor media results in CH_4 evolution and the isolation of crystalline PhC(O)N(Me)ZnMe **57a**.[88] X-Ray diffraction reveals a hexamer composed of *Z*-configured molecules and incorporating two puckered 12-membered $(ZnNCO)_3$ rings, one inverted and staggered relative to the other (Chart 9). Six interactions between the metal centres in one trimer and the carbonyl O-centres in the other yield as many more six-membered ZnNCOZnO rings. The use of more sterically demanding *N*-isopropyl benzamide or benzanilide in tandem with R'_2Zn (R′ = Me, Et) has allowed the structural characterization of tetramers of PhC(O)N(R)ZnR′ (R = ^{i}Pr, R′ = Me **57b**; R = Ph, R′ = Me **57c**; R = ^{i}Pr, R′ = Et **57d**; R = Ph, R′ = Et **57e**).[69,88] These comprise two stacked boat-configured $(ZnNCO)_2$ rings (*e. g.*, Chart 9).[89]



$57a_6$ $57b_4$

Chart 9

The treatment of **57a** or **57b** with ^{t}BuLi gives a product that NMR spectroscopy suggests to be [PhC(O)N(R)Zn(^{t}Bu)$_2$]Li·2thf (R = Me **58a**, R = ^{i}Pr **58b**), a dimeric formulation for which is borne out crystallographically (Fig. 22). At the core of this dimer is a $(LiO)_2$ ring. The Zn centres exhibit distorted trigonal planar coordination and delocalization renders each ligand precisely planar and *E*-configured, which contrasts with the *Z*-configuration noted for Al. However, in common with the Al systems, C$\underline{\cdots}$O and C$\underline{\cdots}$N bond lengths suggest azaenolate character.[90] The reaction of PhC(O)N(Ph)ZnMe with ^{t}BuLi incurs a less straightforward product, and X-ray diffraction reveals an unusual dimer based on the formulation

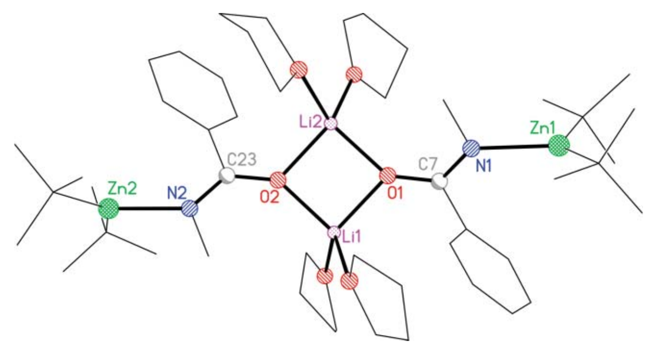
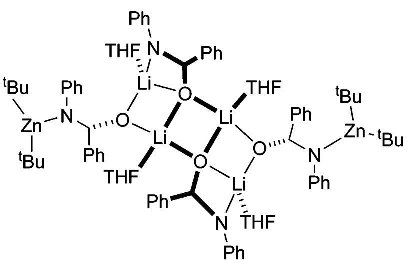


Fig. 22 Molecular structure of the dimer of [PhC(O)N(Me)Zn(^{t}Bu)$_2$]-Li·2thf **58a**.[88]

{PhC(O)N(Ph)Li·thf}·{[PhC(O)N(Ph)Zn(^{t}Bu)$_2$]Li·thf} **59** (Chart 10). This aggregate incorporates a mixture of both lithium zincates and—unusually—*N*-lithiated carboxylic amides,[84b,91] with the combination of *E*-isomerism and coordination of the ^{t}Bu$_2$Zn moiety in the former preventing their *N*-centres from replicating the coordinative behaviour noted for the latter. Plainly, the formation of simple lithium zincates predominates here, and no tendency for hydride formation has been observed. Moreover, the mechanism by which these bis(*tert*-butyl)zincates forms is less straightforward than is the case for their Al analogues.



59$_2$

Chart 10

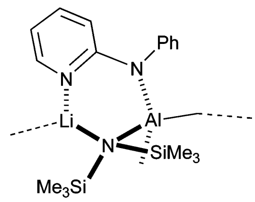


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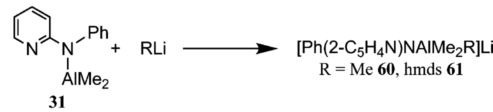
Chart 11



Scheme 9

Probing the mechanism: on the role of the Lewis acid

The Lewis acid appears to play two roles in the formation of hydride containing cages. Firstly, early on in this study it was suspected that the organolithium supplied lithium hydride for entrapment, but that the Lewis acid was required to induce this. Secondly, all of the cages reported above are cationic (either 2+ in the absence of hydride or else 1+ if the interstitial ion is present), and the Lewis acid (Al, Zn, B) is central to provision of the required counter-ion(s).

The inclusion of LiH in these systems can be rationalized simply enough; a charged cluster comprising 6LiAm (Am = amide, guanidinate *etc.*) + 1LiX (X = aluminate, zincate *etc.*) + 1LiH. Logically, given the components present, LiH provision requires a β-elimination process to operate and circumstantial evidence for this has been provided through the reaction of amidoalane **31** with organolithium and lithium amide reagents incapable of β-elimination. In this manner, the simple aluminates [Ph(2-C$_5$H$_4$N)NAlMe$_3$]Li **60** (Fig. 23)[66] and [Ph(2-C$_5$H$_4$N)NAlMe$_2$(hmds)]Li **61** (Chart 11)[60] have been prepared using MeLi and hmdsLi, respectively (Scheme 9). Of course, β-elimination of LiH from ^{t}BuLi results in the co-formation of isobutene, and aliquots of the reaction mixture from which **33** deposits have tested positive for this by GC-MS.[15]

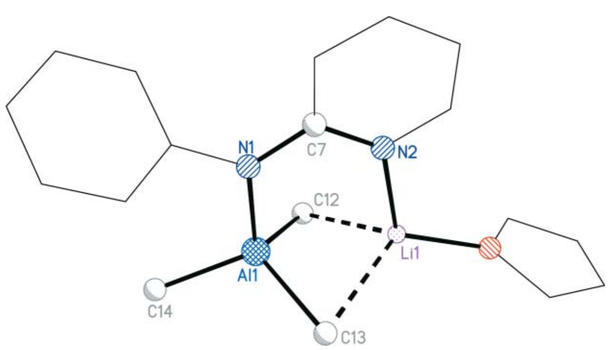
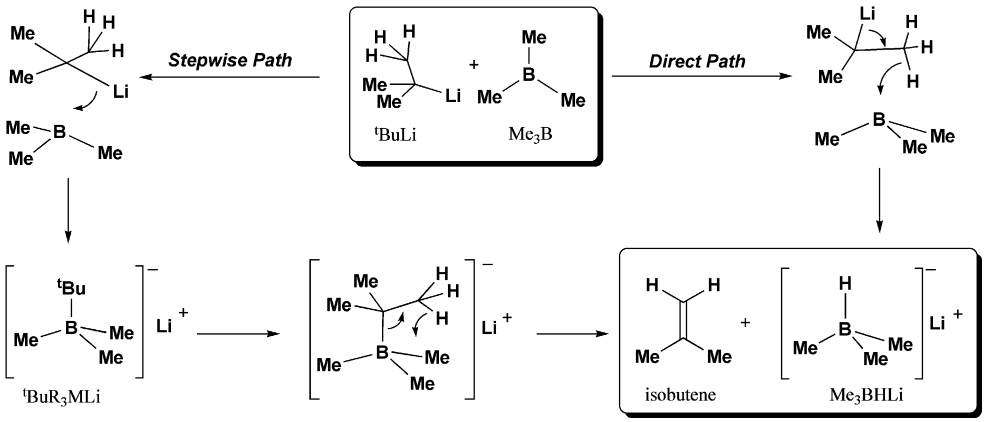


Fig. 23 Structure of the simple aluminate monomer [Ph(2-C$_5$H$_4$N)-NAlMe$_3$]Li·thf **60**.[66]

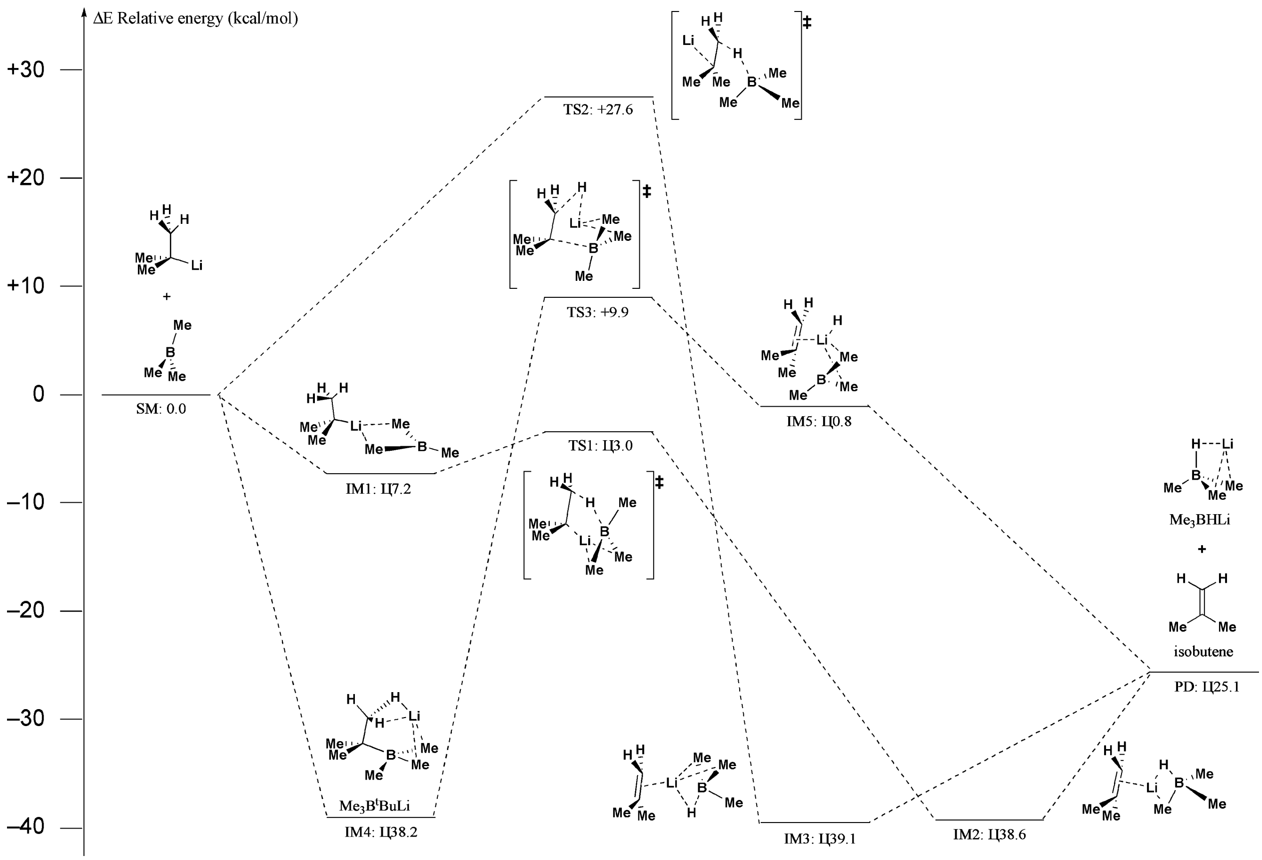
As described above, studies into the ability of Et$_3$B to incur hydride encapsulation have also afforded us a window on the role of the Lewis-acid centres in these systems in aiding hydride elimination from the organolithium substrate. Hence, it has proved possible to observe the borates [Et$_3$B^{t}Bu]Li **62** (by NMR spectroscopy, not to mention incorporated into the crystal structure of **44**) and also [Et$_3$BH]Li (*viz.* tmeda-solvate **47**) as separate materials and 'combined' in oligomeric **45**$^-$.

It was the result of these data concerning boron-based systems, combined with our observation of isobutene production in the synthesis of **33**$^+$, and with the work of Uhl *et al.*, that suggested the ability of alanes to catalyze β-elimination from organolithiums,[92] which persuaded us to probe possible reaction pathways for the Lewis-acid-induced abstraction of LiH from ^{t}BuLi using the model Lewis acid Me$_3$B. DFT calculations (B3LYP/6-31G*) were carried out for two reasons; to establish the viability of an elimination pathway and to investigate the permanence of some of the different anions noted in these systems. Based upon the observation of both [Et$_3$B^{t}Bu]$^-$ and [Et$_3$BH]$^-$ in tandem with hydridic cations (*viz.* **44** and **49**) it was considered that two possible pathways merited study; the initial interaction of ^{t}BuLi and Me$_3$B could lead to the borate [Me$_3$B^{t}Bu]$^-$, which might then undergo intramolecular isopropene elimination to give [Me$_3$BH]$^-$, or else an initial 1 : 1 complex between ^{t}BuLi and Me$_3$B could directly yield LiH through a concerted pathway (Scheme 10). Stepwise reaction followed a barrierless conversion of reagents to give [Me$_3$B^{t}Bu]$^-$ (**IM4**, Scheme 11). However, the high activation energy of +39.1 kcal mol^{-1} required to achieve **TS3** (Scheme 12) renders intramolecular elimination of isobutene unlikely. More plausible were *direct* hydride formation routes. These could involve either (i) the simultaneous transfer of LiH to Me$_3$B (*via* **TS1**) to give an isobutene–[Me$_3$BH]Li π-complex (**IM2**) from which gas irreversibly dissociates, or (ii) direct hydride transfer without the association of Li$^+$ (**TS2**) to give a π-complex (**IM3**) similar to **IM2**. It transpires, however, that this latter process incurs a high activation barrier of +40.4 kcal mol^{-1} (Scheme 12), making the **TS1** route the most probable path to borohydride formation.

All of the cage compounds reported so far have been ion-separates. That is, a mono- or dicationic cluster has been countered by the inclusion of an 'ate anion in the crystal lattice, be it



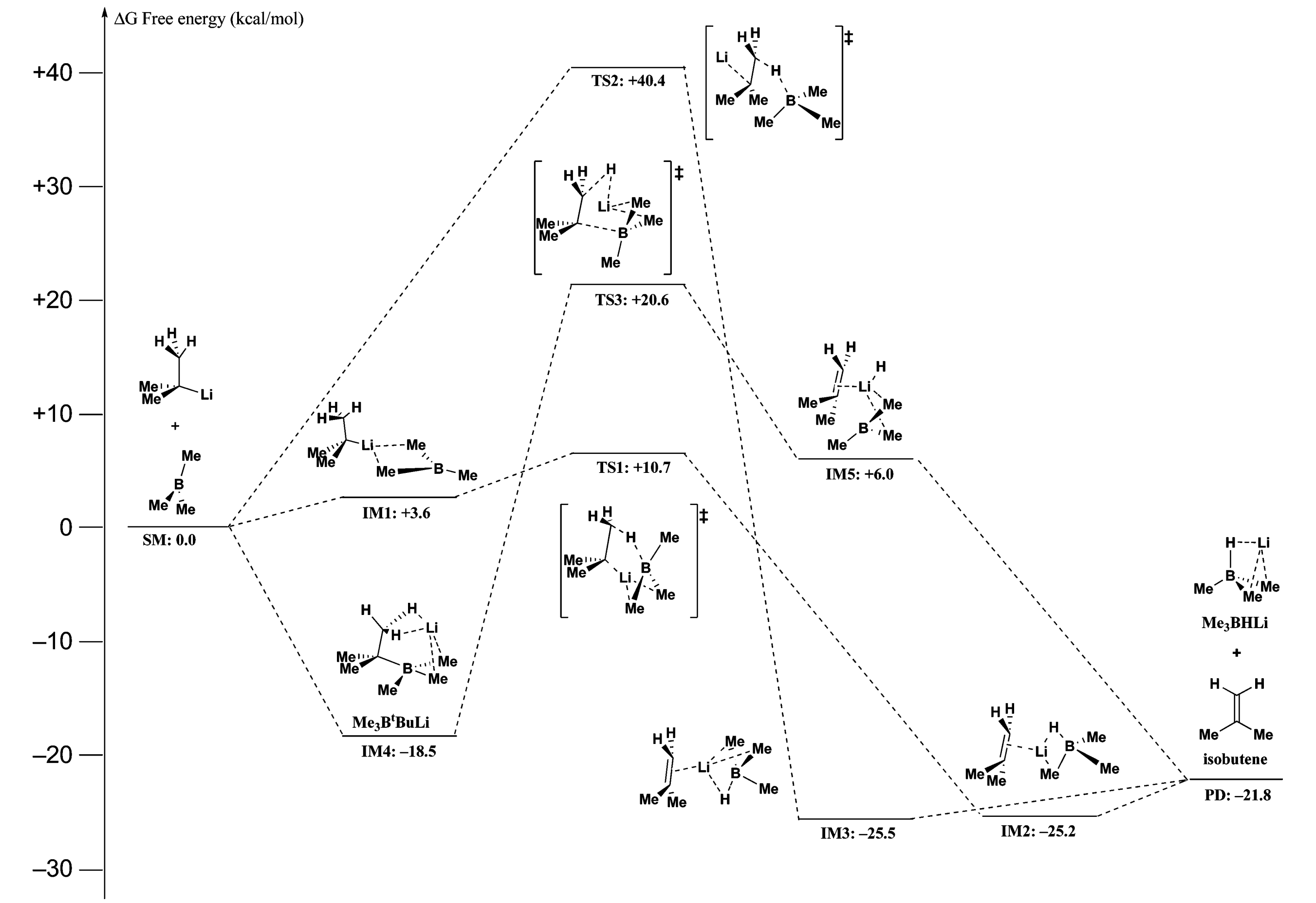
Scheme 10



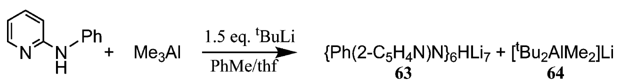
Scheme 11 Possible stationary and transition points during lithium hydride transfer. Energy values (Δ*E*) are relative to **SM** and are shown in kcal mol^{-1}.

aluminate **33**⁻, zincate **39**⁻ or else one of the borates **44**⁻ or **49**⁻. However, it has proved possible to abstract the anion in the presence of a Lewis *base.* This necessarily removes the possibility of the cage carrying a net charge and instead requires the formation of a neutral hydride cluster. In this vein, $\{Ph(2\text{-}C_5H_4N)N\}_6HLi_7$ **63** has been reported. The, albeit low yielding, preparation of this compound (17%, based on aniline) involves the sequential treatment of 2-*N*-pyridylaniline with Me_3Al and ᵗBuLi. On the face of it, this synthesis compares closely with that of **33**. Crucially, however, whereas the ion-separate is prepared using toluene as solvent, **63** is deposited instead (as the only isolable hydridic material) if a toluene–thf mixture is employed (Scheme 13). Conceptually (and, as will be seen below, this *is* a conceptual model rather than an accurate interpretation of the mechanism) Li^+ can be regarded as having been removed from the charged cluster **33**⁺ and to have combined with the ion $[(^tBu_2AlMe_2)_2Li]^-$ (*viz.* **33**⁻) to give, empirically, 2 equivalents of the simple aluminate $[^tBu_2AlMe_2]Li$ **64** as a co-product; evidence for which species comes from ¹H NMR spectroscopic analysis of the reaction mixture.

The structure of **63** is superficially closely related to that of the **33**⁺ ion. However, closer analysis reveals that the absence of one metal vertex in fact introduces some significant differences (Fig. 24, Chart 12).[62,93] Interestingly, the absence of this metal



Scheme 12 Possible stationary and transition points during lithium hydride transfer. Gibbs free energy values (ΔG) are relative to **SM** and are shown in kcal mol^{-1}.



Scheme 13

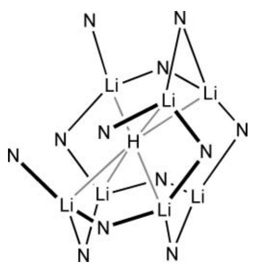


Chart 12

ion allows for subtle differences in the orientations of the organic ligands relative to those in **33**$^+$, and this has implications for our view of the formation of **63**. Three of the six pyridyl N-centres (N3, 3A, 3B) now each interact with just one metal ion (Li1, 1A, 1B) rather than with two. Furthermore, *trans* to the vacant site, Li3 bonds to three deprotonated N-centres (N2, 2A, 2B). The resulting C_3 symmetry cluster is one in which the $(Li^+)_7$ core contains three distinct non-bonding inter-metal distances; those involving unique Li3 (*trans* to the vacant site) are intermediate

Table 6 Bond lengths (Å) and angles (°) in the core of $\{Ph(2\text{-}C_5H_4N)N\}_6HLi_7$ **63** at 160 K[62,93]

Li1–H	1.899(16)	Li3 · · · H	2.49(3)
Li2–H	2.213(15)		
Li1–H–Li2A	176.3	Li1–H–Li1A	102.0
Li1A–H–Li2B	176.3	Li1–H–Li1B	102.0
Li1B–H–Li2	176.3	Li1A–H–Li1B	102.0
Li1A–H–Li2	74.4	Li2–H–Li2A	105.2
Li1B–H–Li2A	74.4	Li2–H–Li2B	105.2
Li1–H–Li2B	74.4	Li2A–H–Li2B	105.2
Li1–H–Li2	78.5		
Li1A–H–Li2A	78.5		
Li1B–H–Li2B	78.5		

(2.586(8) Å) between Li1A · · · Li2 (2.498(7) Å) and Li1 · · · Li2 (2.612(7) Å). The shortest metal–hydride lengths are those between the interstitial ion and the metal centres adjacent to the vacant site (Li1–H 1.899(16) Å) while Li2–H are relatively extended at 2.213(15) Å. This six-coordinate arrangement is severely distorted, with the octahedral motif 'flattened' somewhat, a view most easily recognised if the bond angles about the hydride ion are considered (Table 6). Clearly, while the interstitial hydride retains an approximately octahedral coordination shell reminiscent of that in **33**$^+$, the behaviour of Li3 is not directly comparable to that of the extruded metal ions in the $(Li^+)_8$ system. Whereas,

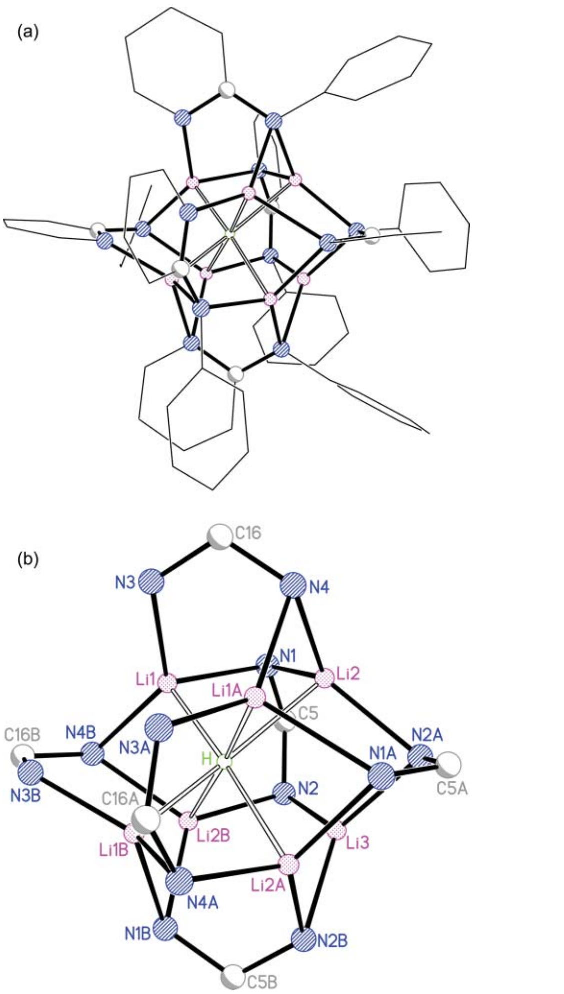


Fig. 24 160 K X-ray crystal structure of the neutral capped-octahedral cluster $\{Ph(2\text{-}C_5H_4N)N\}_6HLi_7$ **63**.[62]

in the latter system, these are withdrawn from the core to the extent that they are clearly non-bonding with respect to hydride (extrusion correlating with the location of charged amido and neutral pyridyl N-centres), displacement of Li3 from the core of **63** is now less pronounced (Li3 $\cdots$ H 2.49(3) Å, *viz.* 2.86(2) Å in **33**$^+$). The hydride is, therefore, best described as occupying a face-capped distorted octahedron.

Combining the demonstrable ability of **63** to form reproducibly with the control gained in hydride encapsulation by opting for reagents capable of supplying LiH and therefore of participating in straightforward hydride translocation, we sought to generalize the formation of electrically neutral hydride encapsulation clusters. Whereas it has already been shown that preformed guanidinate **47** can be treated with Et_3BHLi (3 : 1 equivalents) to give ion-separated **49** (Scheme 6), the synthesis of $(hpp)_6HLi_7$ **65** was targeted by instead using a 6 : 1 ratio. The logic here was that while ion-separated complexes require H^- + $2Li^+$ to be supplied to the cluster (leaving **49**$^-$ and Et_3B which, it transpired, form the Lewis adduct $[(Et_3B)_2H]^-$), a neutral cluster requires that the main-group hydride reagent contributes H^- + Li^+ to cluster formation. In the event, 6 : 1 reaction also afforded an ion-separate crystalline solid, which X-ray diffraction identified as $[Et_3BH]^-[(hpp)_6HLi_8]^+$ **66** (22%, the observation of an ion-separate involving an $[Et_3BH]^-$ anion again makes the hydride the limiting reagent, Fig. 25, Scheme 14).[66] Evidently, therefore, there exists a strong imperative for hydride encapsulation proceeding with 3 : 1 stoichiometry overall, with 2 eq. Et_3BHLi contributing H^- + $2Li^+$ to cluster formation. The (observed) 3 : 1 reaction of **47** with Et_3BHLi proceeds less cleanly than the (idealized) 6 : 1 reaction described above. This is perhaps unsurprising as the stoichiometry of the mixture is now inappropriate. So whereas the formulation of **49** accurately reflects the 6 : 1 composition of the reaction mixture, that of **66** does not reflect the 3 : 1 substrate ratio now in the reaction pot. In contrast to ^1H and ^{11}B NMR spectroscopy on crystalline material isolated from the 6 : 1 system, the presence of significant tetraethylborate **51** is revealed spectroscopically if the product obtained from 3 : 1 reaction is analysed. Whilst the view that combination of a lithium hydride source with an *N*,*N′*-bidentate lithium amide results in hydride translocation is clearly valid, it appears that borate anions which, as has been shown, may take a variety of forms, are sufficiently stable that reaction favours an ion-separated product irrespective of reagent ratios. Thus far it has not proved possible to isolate an analogue of **63**, with all the other hydridic clusters observed (be they μ_2-, μ_6- or μ_8-hydrides) being ion-separates that incorporate a cationic component that comprises a pseudocubic $[Am_6Li_8]^{2+}$ superstructure. Generalization of the chemistry that gives **63** is a target that is actively being pursued at present.

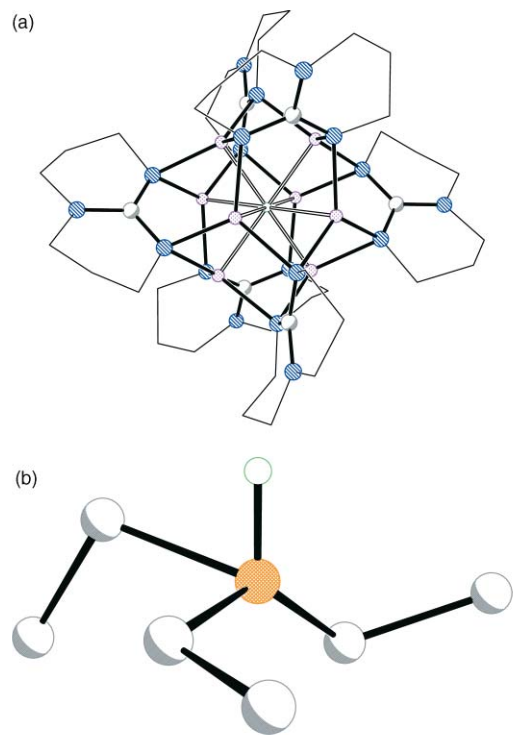
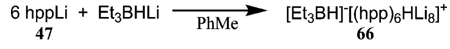


Fig. 25 The X-ray crystal structures of the anion and cation components of $[Et_3BH]^-[(hpp)_6HLi_8]^+$ **66**.[66]



Scheme 14

The isolation of **63** can be taken together with the establishment of how the Lewis acid acts in hydride translocation (*viz.* **49**), to infer that it may be possible begin to manipulate reaction stoichiometry in these systems. Based on experiment and calculation it now appears that formation of hydride-encapsulating clusters can proceed only *via* formation of a boron hydride intermediate and, secondly, that the supplied borane acts pseudo-catalytically, with Et_3B moieties retained throughout the course of reaction (but trapped in various of the ion-separate products). Given the established passivity of Et_3B with respect to aniline substrate in the absence of organolithium reagent and given also that both B–^tBu and B–H interactions exist in product structures, it seems that nucleophilic formation of the borate $[Et_3B^tBu]^-$ is in competition with β-hydride elimination from ^tBuLi under the influence of Lewis acid to give $[Et_3BH]^-$. Similar processes have been noted by others, and $\{(Me_3Si)_2CH\}_m(^tBu)_nAl/^tBuLi$ mixtures have been shown give to $[\{(Me_3Si)_2CH\}_m(^tBu)_nHAl]^-[Li{\cdot}2tmeda]^+$ ($m, n = 1, 2$), whilst in the absence of the amine conversion of ^tBuLi to the hydride and isopropene has been recorded under the influence of catalytic alane.[92]

Based on this, attempts at hydride synthesis employing non-stoichiometric Lewis acid are now being initiated in order to test the view that the p-block substrate does not act stoichiometrically. Given that the utilization of sub-stoichiometric Lewis acid will significantly effect the outcome of reaction (*cf.* the presence of Al, Zn or B in the counter anion to the hydride encapsulation monocation), efforts have been directed towards the synthesis of neutral **63**. It is in this context that the treatment of Me_3Al or Me_2Zn with *N*-2-pyridylaniline and ^tBuLi in the proportions 0.1 : 1 : 1.5 mmol in toluene/thf has been effected with the pure hydride depositing. Isolated yields of 32 mg for Al (18% conversion based on aniline) and 38 mg for Zn (21%, based on aniline), respectively compare favourably with that of 30 mg (17%, based on aniline) obtained when using 1 eq. Me_3Al (*vide supra*). Reaction of Et_3B in a 0.1 : 1 : 1.5 mmol ratio in toluene yields a mixture which X-ray crystallography and NMR spectroscopy reveal to contain **63**. However, the use of toluene–thf instead gives the pure hydride in 27 mg yield (15%, based on aniline).

Conclusions

The various octalithium $[Am_6Li_8]^{2+}$ (Am = amide, guanidinate) structures reported here clearly suggest the viability of superlithiated clusters. Unlike in the reports of $[Li_2]^{2+}$ clusters[94] and $[XLi_2]^+$ (X = Cl, CN, hmds),[95] the presently discussed systems are all highly aggregated cations; the only example of a similarly aggregated lithium-containing cation of which we are aware being that in $[(4\text{-}MeC_6H_4)_6Ta]^-[Br_3Li_4{\cdot}7OEt_2]^+$.[96] Of the compounds prepared by ourselves only one (aluminate **27**) reveals retention of the dicationic octalithium formulation.[55] The remainder all adopt monocationic structures that incorporate hydride, and it has proved possible to establish the generality of this chemistry. Accordingly, $[Am_6HLi_8]^+$ has been revealed with multifarious counter-anions. Initially this species took the form of $[(^tBu_2AlMe_2)_2Li]^-[\{Ph(2\text{-}C_5H_4N)N\}_6HLi_8]^+$ **33**, with its μ_6-interstitial ion established using newly developed neutron diffraction techniques.[97] The coordination sphere of hydride in this example reveals significant similarities to that long-known in the infinite lattice structure of hydride **34**[65] and, in a molecular context, in polyhydride **21**[42] and monohydrides of Ru (**15**$^-$) and Co (**16**$^-$).[34,35] Just as with molecular transition-metal hydrides, variability has been revealed in the nature and extent of hydride coordination by lithium. Indeed, it has been shown that hydride coordination geometry can be both rationalized and manipulated. Thus, a clear correlation can be developed between charge distribution in the ligand and the behaviour of the metal towards the encapsulated ion. Increasing charge localization on the formally deprotonated N-centres in the prerequisite *N,N′*-bidentate ligands yields linearly coordinated μ_2-hydride in **40**$^+$.[66] The bonding pattern in the cluster core correlates precisely with that noted in the *N*-2-pyridylanilide analogue, in the sense that metal–hydride bonding is sacrificed when Li–N$^-$ coordination is highest. Conversely, the distortion in the cores of $[(RR'N)_6HLi_8]^+$ cations can be offset by employing a ligand capable of more even charge distribution. The use of hppH has allowed the characterization of μ_8-hydride coordination in the $[(hpp)_6HLi_8]^+$ cation, countered initially by $[^tBu_3Zn]^-$ in **39**,[66] the field subsequently being extended through the employment of borane reagents (*cf.* **44**, **45**, **49** and **66**). Work with boron has allowed the significant development of our understanding of how the Lewis base component, clearly necessary in the light both of the well established laddering of lithium amides[2] and the work of Uhl,[92] dictates the chemistry of these systems. The observation of $[Et_3BH]^-$-based counter anions accompanying hydride encapsulation (in **45**, **49** and **66**) has suggested the possibility of rationally synthesizing hydride clusters incorporating a counter-ion whose identity can be controlled through the supply of $H^- + 2Li^+$ to cluster formation. It is also hoped to utilize this understanding to apply a monometallic hydride as a supplier of $H^- + Li^+$ and to thereby develop a general route by which to extend the presently limited range of neutral Am_6HLi_7 cluster chemistry (*cf.* **63**).[62] Embryonic studies into whether the Lewis acid acts sub-stoichiometrically in these systems have proved promising, apparently bearing out the view that this is indeed the case. Efforts will next focus on improving the mol% of Lewis acid from 10 mol% to *ca.* 1 mol%. Work is being initiated on the MAS NMR spectroscopy of these systems. Their repeated failure to yield detectable hydridic resonances in solution or simple solid-state experiments (which made neutron diffraction analysis particularly salient) has led us to investigate the possibility of using REDOR methods to probe interactions between metal and encapsulated hydride. Hitherto, the plausible presence of lithium–hydrogen interactions in the counter-anion has suppressed this analytical channel. However, the potential for preparing various boron-centred anions gives us hope that it will ultimately prove possible to gain the control necessary to tailor the anion according to the analytical methods with which we seek to study the cation. Synthetically, having established the need for *N,N′*-chelating ligands, emphasis will shift to focus on building clusters which are peripherally functionalized through the manufacture of suitably substituted ligand frameworks. A further aim is to use our understanding of the role of the hydride provider to probe whether readily available metal hydrides other than those of lithium can be employed to open up pathways to more general heterometallic anion-encapsulation clusters. These will be of both fundamental interest, and also hold out the possibility of multihydride encapsulation.

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